

{ Set Theoretic Solution for the Tuning Problem }

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In this paper I want to suggest a new solution to the problem of musical tuning. On one hand, I see it as a generalization of Just Intonation (JI) to inharmonic timbers, on another, as a unification of spectral interference and harmonicity contributions to consonance within a single framework. The main achievement of the work is the ability to mathematically quantify the phenomenon of musical consonance using set theory. That quantification is done by defining two measures of consonance: affinity and harmonicity. These measures naturally generate sets of intervals that can be used as dynamic tuning systems. The paper is aimed at a broad audience of people who may not be skilled in music and tuning theory or mathematics. Thus, I attempt to give as much details and explanations as I can, while keeping the number of pages as low as possible.

1 Introduction

This chapter is aimed at the reader who is not familiar with tuning theory. Thus, if you know what musical interval is, why it can be represented by a ratio such as $3/2$ and why it is difficult tune a guitar by ear, you can safely skip that chapter. For everybody else, let me try to briefly introduce main concepts of tuning theory and explain why tuning of musical instruments is not a trivial task.

1.1 Basic notions of tuning theory

The most basic notion in music is a notion of a **pitch**. All sounds are perceived in a range from low to high pitched sounds. Example of a low-pitched sound will be a male baritone voice and of high-pitched sound a female soprano voice. The pitch is an absolute position of sound in that range and is directly linked to the frequency¹ with which the sounding body vibrates. Low pitches are produced with low frequency vibrations and high pitches - with high frequency vibrations. The range of all possible pitches is continuous, and in theory, there is an infinite number of pitches.

Since ancient times, people have noticed that if two pitches are sounding together, depending on their relative positions, they may sound musical or unmusical. The relative position of pitches is called an **interval** and can be found as a ratio between frequencies of two pitches. If two pitches are the same (have the same frequency) the interval is described by the ratio $1/1$ and is called a **unison**. If one of the pitches is twice higher than the other (has double the frequency) the interval will be described by the ratio $2/1$ or $1/2$ depending on the order of pitches. For historical reasons the lower pitch is considered the base pitch from which the interval is built. So, the interval $2/1$ is

¹ A detailed definition of frequency is given in the chapter 3.1

called an **octave** and is built up from the base pitch. The interval $1/2$ is also an octave but that is built down from the base pitch. Octave plays very important role in music as pitches an octave apart are considered to be equivalent² (to be the same note). Other prominent intervals are **perfect fifth** $3/2$ and **perfect fourth** $4/3$, together with octave and unison they are the most musical and smooth-sounding intervals.

Given those four intervals we can start combining them to get new intervals. The combination is done by means of multiplication of corresponding interval ratios. For example, to get the interval of perfect twelfth we need to multiply ratios of an octave and a perfect fifth $\frac{2}{1} \cdot \frac{3}{2} = \frac{6}{2} = \frac{3}{1}$. As was noted before, the same interval going in downward direction has inverted ratio. Thus, the interval that is attained by going an octave up and a perfect fifth down is $\frac{2}{1} \cdot \frac{2}{3} = \frac{4}{3}$ a perfect fourth!

Describing intervals as ratios is a classical method that goes back to Pythagoras. Since then, tuning theory seen many changes and nowadays we use a tuning system called 12 tone equal temperament (12-tet) in which intervals are measured in another way. In 12-tet the octave is divided in 12 equal intervals called **semitones** and each semitone consists of 100 **cents**.³ All other intervals in 12-tet are made from whole number of semitones. Perfect fourth is 5 semitones or 500 cents, perfect fifth is 7 semitones, or 700 cents and octave is 12 semitones or 1200 cents. Despite having the same names, 12-tet intervals are different from those defined as simple ratios. For example, perfect fourth as $4/3$ is approximately 498.045 cents, rather than 500 like in 12-tet, and major third $5/4$ is 386.3137 cents and not 400. The only interval that is the same is octave, as 1200 cents give exactly the ratio of $2/1$.

1.2 The tuning problem

Given that some intervals are musical and other are unmusical, it would be very beneficial for composers and instrument builders to have a defined system of musical intervals. Such system is a **musical tuning**. One may notice that defining intervals as musical and unmusical contains a large degree of subjectiveness. That subjectiveness is one of the roots of the tuning problem. By what standard an interval is deemed as musical? If musicality of an interval comes solely from perception, then variance of perception between people will lead to different tuning systems. Perhaps, there exists a natural tuning system that is musical regardless of subjective judgement. If

² That is called octave equivalence. Pitches that are octave apart are different pitches but are the same note.

³ The semitone of 12-tet can be represented as a number $\sqrt[12]{2}$ which is irrational, meaning it is impossible to represent it as ratio n/m .

so, what is such a system? Those are very good questions that baffled theoreticians for thousands of years and that are discussed in detail in Chapter 2. Different answers to those questions lead to different tuning systems. Even within a single European tradition several tuning systems existed alongside each other with some gaining and some losing the dominant status as time moved on. Nowadays, the Western tuning system is very standardized by the use of 12-tet, but there are small communities that preserve traditional tuning systems or experiment with new ones.

Historically, there were attempts to find a natural tuning system that is free from subjectiveness, but the solutions always faced a difficulty in practical implementation. To get a feeling for why, let's see what problem an aspiring guitar player will face when trying to tune a regular 6-string guitar by ear.

Such guitar has six strings, the first string has the highest pitch, and the sixth string has the lowest pitch. The standard tuning has the interval of major third between second and third strings and the interval of perfect fourth between all other pairs of neighboring strings

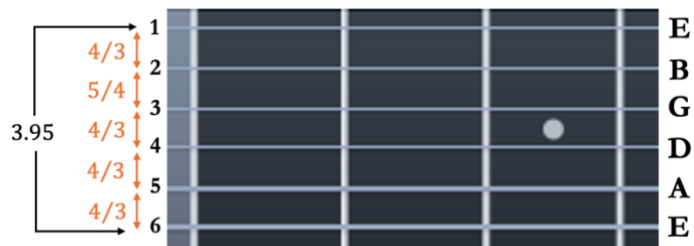


Figure 1.1 - Demonstration of the problem of tuning of 6-string guitar to just ratios. Adding up the intervals between neighboring strings (in orange) creates the dissonant interval of approx. 3.95 instead of 4/1 between low and high E.

(Figure 1.1). When tuning all those intervals by ear, we naturally pick out 4/3 ratio for perfect fourth and 5/4 ratio for major third. Even small deviations from those exact ratios are perceivable for the careful listener. In the properly tuned guitar, the first and sixth strings should be exactly two octaves apart with ratio 4/1 and thus form a perfect consonance. However, that is not the case in the way we tuned the guitar, and the reason for it is purely arithmetic. The ratio between first and sixth strings is the product of multiplication of intervals between all other neighboring strings, so from low to high strings it will be $\frac{4}{3} \cdot \frac{4}{3} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{4}{3} = \frac{1280}{324} \approx 3.95$, which is a little bit short of 4/1 and sounds noticeably out of tune. Moreover, almost every chord on such guitar will sound a little bit off, which renders the instrument unplayable. The solution to that problem is not straightforward, as it depends on our values, goals and subjectiveness of our perception.

That is the heart of the tuning problem. The natural intervals to which our senses gravitate to are difficult to use in musical instruments and the subjective factor in the perception of musicality of intervals further complicates the ways this issue can be fixed. In following chapters I will suggest my solution to that problem. Along the way, we will delve into philosophy, mathematics, physics and psychoacoustics. The long and complicated journey will result in a mathematical description of consonance and a candidate for a natural tuning system.

2 History and methodology of tuning theory

In this chapter, I want to share my view of the historical and philosophical context that surrounds the tuning problem. I attempt to briefly describe the historical development of the main traditions of thought within that field. Doing that allows us to see a larger picture in which modern tuning system is one way among many to solve the tuning problem. Given the empirical knowledge and technological possibilities of XXI century, it can be argued that 12-tet has lost its relevance in musical practice. Thus, it might be time to consider other schools of thought and their influence on music and tuning. My analysis is scoped for European musical traditions that stem from Ancient Greece and does not include music of other civilizations. The aim of that chapter is to explain my motivation for the current work and to describe philosophical and methodological foundations behind it.

2.1 Pythagoras, Ptolemy, Aristoxenus and early Modernity

The word consonant comes from Latin **consonantia** "harmony, agreement". It is directly linked to the musicality of intervals that was discussed in the Chapter 1.2 and describes an observation that some intervals sound smooth, stable and harmonious, as if they fit together forming one larger whole. The antonym of consonance is dissonance, a sensation of pitches clashing, not "agreeing" with each other, sounding rough and unstable. It becomes apparent when delving into the field of musical tuning theory that consonance is a central phenomenon around which different tuning systems are built. Thus, the greatest factor that influences the construction of tuning theory is our conception of the nature of consonance. For millennia, people realized that consonance can be judged by consulting our senses or by invoking reason⁴.

The most ancient formulation of a musical tuning system in Europe was done by Pythagoras in Ancient Greece (VI century B.C.). There is no direct text that was left by Pythagoras himself, so his teaching is known to us by the works of his followers called Pythagoreans. They believed that consciousness precedes matter and that material phenomena are manifestations of timeless Ideas. The famous saying that is attributed to Pythagoras "All things are Number" means that all material phenomena are measurable and can be represented as numbers. In that sense, musical pitch can be expressed as how often the musical body beats the air per unit of time or what we now call a

⁴ [15] Book 1, p.3: "The criteria in harmonics are hearing and reason, but not in the same way; hearing is the criterion for matter and condition, while reason is the criterion for form and cause."

frequency. Consequently, musical interval being the ratio between two pitches, can also be represented as a number. In a Pythagorean view, those numeric relationships are what really define consonance and reasoning about those relationships gives a theory of musical tuning. Pythagoreans concluded that ratios $1/1$, $2/1$, $3/2$ and $4/3$ define perfect consonances of unison, octave, perfect fifth and perfect fourth accordingly and therefore were placed in the basis of their tuning theory.⁵ The fact that small deviations from those ratios, such as $41/30$ for perfect fourth also sound quite consonant to our ears was seen as a flaw of auditory system that is incapable of perceiving such small distinctions. Pythagoreans compared judging consonance by ear alone to judging distance by eyesight alone.

The opposite view, that relied on senses more than reason to judge consonance, also emerged in Ancient Greece and was formulated by Aristoxenus (IV century B.C.). He was one of the most talented students of Aristotle that produced more than 400 works on different topics. In his work on music “Elements of Harmonics” he opposes Pythagoreans and gives preference to senses as a main judge of consonance.⁶ If a combination of notes is perceived as consonant, it is consonant. The role of reason in such case is to serve senses by helping musical practitioners in reaching their goals. Instead of using numeric ratios to represent intervals, Aristoxenus suggested to measure musical intervals as a sum of discrete quantities. That approach is very similar to the modern approach of measuring intervals in cents.

In II century A.D., in his treatise Harmonics, Ptolemy criticizes approaches of Pythagoreans and Aristoxeneans and suggests his own vision of the role of reasoning and senses in judging consonance. Ptolemy shares idealistic view of Pythagoreans that mind precedes matter but argues that senses play a large role as well, even given their limitations. He argued that it is unreasonable for some Pythagoreans to dismiss senses when they profoundly disagree with their theory. When the contradiction with theoretical prediction is undeniable, the theory must be reconsidered. In his view, true theory should be in harmony with both reason and senses.⁷

Those are three main schools of thought in tuning theory that I believe are still relevant to this day. They can be put into two categories. On one hand, we have Pythagorean and Ptolemean theories that give precedence to the conciseness and the world of Ideas and therefore are **idealistic**. On another hand, we have **empirical** theory of Aristoxenus that emphasizes the role

⁵ [15] Book 1, chapter 1.5.

⁶ [16] p. 196: “There is a certain marvelous order which belongs to the nature of harmony in general; in this order every instrument, to the best of its ability, participates under the direction of that faculty of sense-perception on which they, as well as everything else in music, finally depend.”

⁷ [15] Book 1, chapters 1.6 - 1.9.

of senses and the material phenomena. Both types of theories employ sensory observations to inform the mind about the reality, but they differ in what type of reasoning is used.

In idealistic theory, reasoning is a combination of abductive and deductive reasoning. Characteristic feature of such reasoning is top-down approach. At first, top level axioms or key Ideas of the theory are guessed using intuition (abductive reasoning), and then, the consequences of those ideas are obtained using deduction. Pythagoras guessed that all things are numbers and that numbers 1, 2, 3 and 4 are sacred⁸, thus concluding the consonance of 1/1, 2/1, 3/2 and 4/3 as all possible ratios between those numbers (not counting inverse ratios). Knowing that, one can use deduction to find other intervals in tuning. Pythagoreans used the process of stacking and combining consonant intervals to get their tuning system.⁹ Hard idealistic theories like Pythagorean theory downplay the role of empirical and sensory argumentation while soft idealistic theories like Ptolemean theory seek to find harmony between reason and observations. Thus, in soft idealistic theory the deduction should be used to find testable predictions of initial axioms to validate the theory. It may come as a surprise, but most if not all great theories in physics are soft idealistic theories. Einstein guessed the constancy of the speed of light, which was later confirmed by observations, not the other way around. Similar guesses were done by Newton, Maxwell, Heisenberg and other great scientists in their famous theories.

Empirical theories, such as Aristoxenian theory, are built from bottom-up using inductive reasoning. At first, the collection of observations is gathered that can be used to formulate low level ideas and principles. Those principles can then be further inductively generalized until sufficient explanation power is reached and the body of the theory is deemed completed. Such theories are frequently employed in social sciences and can be very effective as vehicles of reaching practical goals. But gaining fundamental knowledge about the structure of the universe, seem to be done by idealistic theories.

Historically, in the West, idealistic view of consonance was dominant up until early Modernity when Renaissance thinkers turned to ancient texts in an attempt to resurrect the lost music of antiquity. The XVI century debates between Gioseffo Zarlino and Vincenzo Galilei (the father of Galileo Galilei) provide an insight into the philosophical split that happened within the field of

⁸ [17] “Whole number in the view of Pythagoreans had a sacred meaning. Number 1 represented a unity and origin of all things, number 2 symbolized female principle, number 3 male principle and number 4 symbolized justice. Number 10 that is a sum of numbers 1, 2, 3, 4 was most perfect and meant unity that arises from multiplicity. All 5 numbers formed a tetractys the holy object that was worshiped by Pythagoreans.”

⁹ [8] Chapter 4. The full list of intervals of Pythagorean tuning: 1/1, 256/243, 9/8, 32/27, 81/64, 4/3, 729/512, 3/2, 128/81, 27/16, 16/9, 243/128

tuning theory.¹⁰ At that time, the most common tuning system was Pythagorean, in which intervals of thirds were not considered to be consonant¹¹. However, senses of everyone who attended church services and listened to chorales perceived them as consonant intervals. That was a problem, as it meant that chorales were not using intervals of Pythagorean tuning, but something else. Another problem was the application of Pythagorean tuning to fretted and keyboard instruments. Instruments tuned in such way were not suitable for performing triadic harmony and sounded out of tune in many tonalities.

Gioseffo Zarlino set out to overcome those challenges by formulating a new tuning theory which earned him a status of one of the greatest musical theoreticians of his time. Zarlino worked within established idealistic tradition of thought and turned to Ptolemy for the explanation of triadic consonance. He reasoned that Ptolemy's intense diatonic scale, also known as the Ptolemaic sequence¹², was the natural tuning that was fundamental part of the reality, independent from anyone's perceptions or point of view. The major third in that tuning was represented by much simpler ratio of $5/4$ than Pythagorean $81/64$ and sounded much smoother. Zarlino argued that natural instruments like human voices automatically gravitate towards Ptolemaic sequence when singing polyphonic music. Artificial instruments, like keyboard or fretted instruments, have fixed positions for notes and thus are incapable of producing correct Ptolemaic Sequence in all tonalities. Therefore, as a sacrifice to make those instruments sound better, we must artificially temper the intervals we tune them to, using reasonable theoretical method.¹³

Galilei was a student of Zarlino, but being a lute builder, he was much more practically oriented than his teacher. Combining that with a counter-cultural stance it sent Galilei on a very different path on tackling the same theoretical issues. Galilei wrote several responses to the works of Zarlino, where he argued that there is no such thing as natural tuning and that all tuning systems

¹⁰ For the detailed analysis of the Galilei-Zarlino dispute refer to [11]

¹¹ Pythagorean tuning is obtained by stacking perfect fifths on top of each other (raising $3/2$ to the n -th power) and transposing the result back to be within the range of an octave (between $1/1$ and $2/1$). The interval of major third obtained in such way is $81/64$ and minor third is $32/27$, which do not sound very pure to the ear. The Pythagorean tuning for the major scale consists of the following intervals: $1/1$, $9/8$, $81/64$, $4/3$, $3/2$, $27/16$, $243/128$, $2/1$.

¹² Ptolemy's intense diatonic scale also known as Ptolemaic sequence or 5-limit just major scale, consists of the following intervals: $1/1$, $9/8$, $5/4$, $4/3$, $3/2$, $5/3$, $15/8$, $2/1$. Note that it shares a lot of intervals with Pythagorean major scale, but ratios for major third, major sixth and major seventh are much simpler.

¹³ In Ptolemaic sequence the intervals are not evenly spaced within the octave, so performing the same melody or a chord built from different notes will contain different intervals. Tempering shifts intervals from their just values in order to make a tuning more even. From idealistic perspective, tempering artificially redistributes dissonance across tuning. Very dissonant intervals become less dissonant at the price of consonant intervals becoming less consonant, thus this process should be done carefully, guided by a reasonable theory.

are constructs of a theoretician¹⁴. He also argued that it is not correct to divide instruments on natural and artificial, as choirs and instruments that accompany them are fundamentally equal and perform in the same tempered tuning. He referred to Aristoxenus as an authority to argue a primacy of senses in music and necessity of tempering intervals to the values established by sensory observations.¹⁵

Both Zarlino and Galilei argued in favor of tempering intervals, but from different perspectives. For Zarlino it was a sacrifice to make artificial instruments playable and that tempering should be theoretically justified, while for Galilei it was obvious solution to the practical problem, and one could use experimentation to derive the correct tuning. With time, the design of keyboard instruments was perfected. The great convenience that was provided by keyboard instruments, made them must have instrument for any serious music practitioner or a composer. Early tempered tunings were made to minimize tempering and to keep as many thirds pure as possible adhering to Zarlino's worldview. At the same time, the entirety of music theory, notation and tuning started to form a self-contained system of music that standardized a musical language, education and composition. To perfect the whole system of music, the tuning had to be further tempered.^{16,17} In modern 12-tet system all intervals apart from octave are tempered, which from idealistic perspective means they are dissonant.

2.2 Modern consonance research

Nowadays, tuning theory is rarely taught in music schools and 12-tet intervals are just given to students as a fact of reality of music without any explanation. As a result of gradual shift towards empiricism and materialism that happened since XVI century, the consonance of intervals is explained by appeal to senses. If a musician desires to delve into the reasons of why intervals are the way they are, he or she will be presented with the evolutionary narrative. In such view, tuning theory first emerged as the Pythagorean system that was later improved by various meantone temperaments and that in early XX century reached its perfected state in modern 12-tet system.

¹⁴ [11] p.5: "Galilei demands that we question the assumed 'naturalness' of musical scales or tuning systems. In his view, there was no justification to conclude that Ptolemy's Syntonic tetrachord was any more natural than those attributed to Pythagoras, Archytas, or Didymus: all were human constructions and thus were 'artificial.'"

¹⁵ [11] p. 82: "Galilei then justified the tempering of individual tones and semitones through the authority of Aristoxenus and explained its application through a practical demonstration on the keyboard."

¹⁶ [12] Chapter 17 "Equal temperaments".

¹⁷ In older tempered tunings such as meantone tuning the thirds were tried to be kept pure. That lead to different tonalities having different color or flavor, as their triads were slightly different. Some tonalities were outright unusable because of very dissonant triads. To remove that problem and allow for free modulation and transposition, a further tempering had to take place that sacrificed the purity of triads leaving only the octave untouched.

The merit by which 12-tet is judged as superior system is the ability to freely modulate between different tonalities and its suitability for fretted and keyboard instruments. Natural preference towards simple ratio intervals is downplayed in favor of convenience, to the point that tempering is not thought of as sacrifice, but rather as natural and necessary procedure to make music that sounds in-tune. That narrative provides an illusion of completeness and is very convincing as it is in tune with a broader philosophical worldview and everyday experience. We all listen to 12-tet music that sounds just fine.

If a musician wishes to understand the origin of consonance or what is that they hear as consonance or dissonance, there is a vast scientific literature on that topic. Such research is carried out on the intersection of music theory, acoustics, biology and psychology. The methodology is purely empirical and relies on biological facts about auditory system and behavioral studies as a source of knowledge about consonance. A good review of modern state of research can be found in [1]. In this article, authors describe 3 predictors of consonance that can be used to categorize all tuning theories into 3 families. Those predictors are cultural familiarity, harmonicity / periodicity and spectral interference.

Theories based on cultural familiarity propose that there is no natural phenomenon that defines consonance, perception of consonance is fully subjective. The musical tradition of a culture forms a consensus among individuals and conditions them to perceive familiar music as consonant and foreign music as dissonant. This thesis relies on two observations. The first one is that a set of consonant intervals varies substantially between different cultures. The second one is that every human being has a natural innate preference for the things that are familiar instead of things that are alien. Thus, the music that you are brought up with will sound more pleasant and that sense of pleasant familiarity is what we call consonance. That way, tuning is completely arbitrary with no grounding in objective reality.

Periodicity and harmonicity theories suppose that our auditory system evolved to recognize harmonic (periodic) sounds from the environment. Discerning such sounds is important for our survival, as they are usually produced by humans and animals. The characteristic feature of harmonic sounds is that they have a pitch. The easier for our auditory system to determine that pitch, the more consonant the sound is. Periodicity and harmonicity theories are less subjective than cultural familiarity theories as there is an objective mechanism with which waves of different pitches interact to form a single pitch (more on that in Chapter 3.3). That mechanism does explain the natural preference for simple ratio intervals and our ability to tolerate small deviations from those ratios. Because the origin of consonance is sensory, the exact value for consonance of the

particular interval depends on the mechanism with which our auditory system detects a virtual pitch. Currently there is no consensus on what that mechanism is, but roughly one can say that predictions of harmonicity / periodicity theories conform with conventional music theory. There is, however, a set of observations that this family of theories cannot explain. One important example, is tuning of inharmonic sounds.¹⁸

The last family of consonance theories is spectral interference theories. They are based on the fact that each real sound is made of elementary vibrations called partials.¹⁹ When several sounds are present together, their partials interact with each other in our inner ear. Those interactions can cause the sensation of dissonance. An important example of spectral interference theory is the work of William Sethares “Tuning, Timbre, Spectrum, Scale”. Sethares builds upon the idea of Helmholtz [2] that dissonance can be defined as a rough beating²⁰ between partials of different sounds. Consonance in that sense is lack of such rough beating. Sethares introduces an empirically derived characteristic of human auditory system called dissonance curves that quantifies the amount of dissonance of a particular sound. That characteristic can be used to derive a tuning system. The predictions of his theory fit well with common music theory, as well as observations of tuning of inharmonic sounds. Sethares demonstrates that by predicting the tuning for the Gamelan²¹ orchestra from the spectrum of individual instruments. The most exciting part is that Sethares theory has a great practical potential, as it provides a vision for new instruments with dynamic tuning and inharmonic timbres. It has its limitations though, as it does not provide a meaningful tuning for sounds with small number of partials. But most importantly, it does not explain why inharmonic sounds may sound dissonant even when no rough beating is present.²²

All three families of theories have solid reasoning behind them, but notice, that all of them are empirical theories that place the origin of consonance in human auditory system and perception. Thus, we can conclude that at present day, empirical and Aristoxenean worldview is dominant

¹⁸ Example of inharmonic sound is a sound of church bell. If the sound sounds “metallic” it is probably inharmonic. The issue with such sounds is that conventional consonant intervals, such as octave, may sound dissonant while deviations from those intervals may sound consonant, as was demonstrated by William Sethares in [4]

¹⁹ Partial is an elementary sine wave vibration. A complex sound is made of many partials with different frequency. Mathematically, partial is a prominent sine wave component in Fourier transform representation of the wave.

²⁰ Beating is the phenomena that is a product of interference of two waves with close frequencies. For explanation refer to [4] Chapter 3 “Sound on sound”.

²¹ Gamelan is the name of the traditional Indonesian music that utilizes highly inharmonic metallophones. The tuning of Gamelan varies from orchestra to orchestra and significantly differs from common 12-tet. That difference is so huge it contradicts all common European knowledge of consonance of intervals. Sethares theory is capable of explaining the tuning of Gamelan and the variation of that tuning from orchestra to orchestra.

²² Sethares involves cultural familiarity argument to explain why harmonic sounds are preferred to inharmonic sounds and that it takes time to get used to inharmonic harmonies. However, he also admits that ear tends to reset after some time to prefer harmonic sounds once again.

within the tuning theory. Having said that, idealistic and Pythagorean tradition is not gone. It is alive today within microtonal community of musicians and is known as Just Intonation (JI). JI is an octave based tuning system that is built from intervals represented as simple numeric ratios. It is commonly accepted within that theory, that the degree of consonance of an interval is inversely proportional to the size of the numbers used to define that interval.²³ This way the Ptolemean third of 5/4 is more consonant than Pythagorean third 81/64 as smaller numbers are used in its ratio. Every rational number is to some degree consonant, and the historical development of tuning is seen as a slow process of recognizing that truth²⁴. Modern 12-tet tuning relies on irrational numbers for intervals such as multiples of $\sqrt[12]{2}$, which are dissonant from JI point of view. The fact that for many modern listeners 12-tet third will be preferable to Pythagorean third can be explained as an imperfection of our hearing and perception.

2.3 The hierarchy of musical interfaces

Neither discoveries made by scientific consonance theories nor the opposition in the face of JI were capable to significantly diminish the popularity of 12-tet tuning system among practicing musicians. I believe that there are 3 reasons for that: network effect, heritage and practicality. Vast majority of musicians speak the language of 12-tet, which provides huge benefits to the community, as it enables communication across cultures and genres. Moreover, 12-tet gives access to the incredibly rich library of masterpieces from church chorales to modern electronic music. One can make a case that music prior to early XX century was not written in 12-tet, thus performing Beethoven or Chopin in 12-tet is not historically accurate.²⁵ However, 12-tet does a good enough job approximating different meantone temperaments, so there is no historic piece that is unplayable in 12-tet and the difference in sound is only perceivable for a skilled listener. That naturally brings us to the third point that 12-tet is very handy as a practical tool. Since XVI century onwards, as philosophical foundations started to become more and more materialist, practicality became a very persuasive argument that is responsible for popularity of tempered

²³ [8] Chapter 1 “One definition of consonance – the one used in this book – is that two pitches are consonant when their vibrations can be related by small numbers”.

²⁴ [12] Chapter 15: A thumbnail sketch of the history of intonation

²⁵ In meantone temperaments all tonalities had their own color, as intervals were not uniform as in 12-tet. That is the reason why great composers of the past choose different tonalities for their pieces. Sometimes that choice of tonality leads to more difficult fingering on an instrument, thus demanding a greater skill of the performer. In 12-tet, that color difference between tonalities is lost. So, the only difference that remains is the relative pitch difference and different fingering.

tunings. If there were something more practical than 12-tet, or something desirable that 12-tet could not replicate, then 12-tet would not get the reach it got.

The practicality of 12-tet is twofold. One benefit is that it suits very well for fretted and keyboard instruments, in fact the whole process of tempering was introduced to tuning theory in order to make those instruments sound in-tune. The second benefit is that it simplifies the language of music. Modern musician spends countless hours learning to read sheet music, what notes work together and how to express certain feelings and emotions with music. Thus, any simplification on that journey is very welcomed. Meantone tuning made a modern keyboard possible, the layout of white and black keys is a marvel of engineering. It is easy to navigate and comprehend, and it gives so many possibilities to a skilled performer that it became the go-to tool for every composer. The shift from meantone to 12-tet removed expressive nuance of different tonalities but at the same time it made the language of music more coherent and gave new possibilities to experiment with modulation.²⁶

The alternative tuning systems fall short in delivering better practicality than 12-tet. In fact, they add a lot more complexity to the already very complex art of music. Let's consider JI as an example. Instead of learning 7 notes, musician will need to learn different flavors of those notes and start dealing with arithmetic of simple ratios. Those different flavors must be somehow written down in the notation, so now the complex art of sheet reading becomes even more complicated. Moreover, extra difficulty is added to the instruments used to perform JI music. Keyboards that can approximate intervals of JI are not widely available and using fretless instruments requires even greater skill and ear training to hit the correct notes consistently. Only the brave souls take that route, and they do reap the benefits of a unique sound of JI that is not available for 12-tet musicians.

Having said that, I believe that this critique may not be as relevant as before. Modern digital technologies can simplify most of those troubles. Nowadays, people listen to the records rather than live shows. Such music is either recorded in controlled studio environment or entirely digitally produced. That lowers the skill necessary to perform music outside of 12-tet. Digital notation is vastly different from classical notation and majority of contemporary musicians do not know or need to know classical notation. Notes in modern software is represented as a grid of MIDI events on the screen known as piano roll. It seems plausible that flavors of notes can be much more easily

²⁶ Modulation is the process of switching from one tonality to another. Different transitions have their own specific character that can be used to express certain feeling. In meantone tunings some tonalities are unplayable, thus one cannot modulate to them. In 12-tet there is no limitation on modulations as all tonalities are equal.

encoded on the screen, than on the piece of paper. Moreover, computers can do necessary mathematics for the musician removing the burden of dealing with numbers.

We can conceptualize instruments and notation as an interface to tuning, which is itself an interface to music. In the case of 12-tet, piano keyboard and classical notation are interfaces to 12-tet, which in turn is an interface to European music. A lot of times you can substitute one of those elements without a need to change the other two. For example, you can substitute piano with guitar and notation with tabs and keep 12-tet and European music intact. One can think of many similar examples, however, as we get up the interface hierarchy it is getting more difficult to keep something intact. For example, a small change from 12-tet to meantone temperaments does tolerate keeping keyboard and notation, but it limits the scope of European music as some tonalities are inaccessible in meantone tuning system that are accessible in 12-tet. And if you change the European music to traditional Middle Eastern music, you must consider switching both tuning and instrumentation to have a faithful performance.

The great thing about 12-tet is that it can approximate a great variety of music. It can even approximate the music of other cultures such as Middle Eastern music, but with varying degree of success and with inherent Europization of sound. Sometimes, however, the music is just too alien for 12-tet. There are attempts to perform Gamelan in 12-tet but in my opinion it does not resemble real Gamelan at all. That property of 12-tet as king approximator has to be recognized and put in its proper place, it is a tool to reach a certain sound and provide a certain utility, but it is not a universal truth of music.

Having that mindset, we can start thinking about tuning as a tool that serves us. In some cases, the distinct stable sound of JI would better suit a composition than 12-tet. The problem that holds many musicians away from experimenting with tuning is a lack of proper interface (instrument and notation) to tunings other than 12-tet. Still to this day, piano keyboard is a dominant tool for composers, and it inherently leads to 12-tet. That coupling of keyboard and 12-tet was noticed by Harry Partch in the middle of XX century, which set him for a quest of building new musical instruments that interface to a different tuning and produce different music. The same coupling exists between 12-tet and modern notation that powers all electronic music – the MIDI protocol. By design MIDI was developed as a digital interface to 12-tet specifically. Thus, for the 12-tet to get its fair place as a keyboard specific interface to European music and for other tunings to become relevant, a new interface to tuning has to be made. That interface should be practical and allow for effortless experimentation with tuning. Even better if that interface will be simpler to use

that 12-tet and piano keyboard when accessing European music. But what tuning theory should such interface target, if any?

2.4 Methodology for a new theory

The question if natural or non-subjective tuning exists or not, baffled theoreticians for thousands of years. It is of primary importance when building an interface to music. If natural tuning does not exist, one is free to choose any tuning to suit their needs. In that case an interface to tuning should be as flexible and non-discriminating as possible so that it does not influence musicians' choice for a particular tuning. Digital technologies are flexible enough to support such an interface. However, I am firm believer that natural tuning exists. That belief comes from me believing in the benevolent God and the intelligent design of Creation. Furthermore, I believe that such tuning can be found and is expressible in the language of mathematics. That belief is what some call Pythagorean faith [3]. For those who do not share my beliefs, I think that success of Sethares theory to describe the music of Gamelan and European music within a single theory is indicative of the existence of natural tuning. How else such prediction is possible if there is no cross-cultural, non-subjective mechanism that determines tuning. However, in final analysis, one has to take the existence of natural tuning on faith.

If natural tuning exists, the interface to music should be built around it. One can search for such theory in two ways. First one is empirically, by collecting as much data as possible and then inductively generalizing it up to a set of ideas that describe all empirical data. The work [1] is a good example of such approach as it starts by generalizing empirical findings into three mechanisms of consonance perception²⁷. And then determining a proportion each mechanism play in our perception of consonance to best fit the empirical data. That approach works, but it has a major problem. It does not suggest natural tuning system. No matter how much empirical data you gather, there always will be some things that contradict it. You will have to always tweak some parameters to improve the system. Such approach can only inform the mind on where to look for natural system, what to pay attention to, but it will never find it.

The only way to find the natural tuning is by suggesting an idealistic theory. Idealistic theory places origin of consonance and thus tuning outside of material, sensory realm. It starts by abductively postulating a set of axioms from which the rest of the theory and testable predictions are derived

²⁷ Cultural familiarity, spectral interference and harmonicity / periodicity.

deductively. The benefit of such theories is that their axioms are usually very simple mathematical ideas that are timeless and can be transmitted across millennia. To claim validity of the theory, one can choose the way of Pythagoreans, postulate your reasoning and claim the correctness of the theory by reasoning alone. Or the way of Ptolemy, to judge correctness by how well predictions of the theory fit the empirical data. I believe that Ptolemean methodology is the same that was used in physics from the times of Newton and that is responsible for the staggering achievements in the description of the material reality.

As we discussed before there is no modern theory of consonance that describes all empirical data. In table below one can see what is missing in each family of theories.

Table 2.1 – Limits of applicability of current tuning systems

	Conventional harmony	Consonance of inharmonic sounds	Provides tuning system for pure tones (sine waves)
Just Intonation	✓	✗	✓
Harmonicity / periodicity	✓	✗	✓
Sethares spectral interference	✓	✓	✗

The natural tuning theory should check all the boxes and hopefully has testable predictions outside of what is stated here. Such theory is free from subjective factors as consonance is not sensory in its origin. Senses detect consonance as an eternal mathematical reality and inform the mind about it. The mind, depending on its state can have different disposition relative to that reality by admitting or rejecting it.

In the past, attempts of formulating such theory were done by using the Idea of proportion as the one responsible for consonance. The same way our eye recognizes the beauty of correct

proportions in architecture or visual arts, our ear recognizes in sound. That beauty of properly proportioned sounds is what is perceived as consonance. That is the basis for Pythagorean, Ptolemean, Zarlino's and JI tuning systems. In current work, I suggest another mathematical approach that can lead to natural tuning theory. I propose that consonance is not codified in mathematics of proportions, but rather in mathematics of sets. Using that approach, we can quantify a value of consonance for a given interval that accounts for both spectral interference and harmonicity within a single framework. That makes me assume that proposed theory checks all the boxes in the Table 2.1, though I do not perform a study to prove that.

I believe that proposed approach is promising enough to be implemented in practice. However, I do not suggest that proposed theory is a natural tuning system in its perfected state. More research and critical analysis have to be carried out. My intuition tells me that there is a better, more general mathematical idea behind consonance than the one proposed in this work. I share my thoughts and conclusions as a possible steppingstone to inspire others to find what I can't, or to use what I discovered by making some good music.

3 Mathematical representation of sound

Conventional tuning theories are built on the assumption that musical sounds have a pitch. Such sounds can be characterized by the frequency of the pitch. Thus, a single number is enough to represent a musical sound. In reality, sound is much more complex phenomena and to capture that complexity, we have to find a more appropriate mathematical representation.

3.1 Physics of waves

From physical standpoint, sound is a longitudinal pressure wave that propagates in a medium, usually air. To create such a wave a body has to hit a medium to create a zone of high pressure that will propagate with constant speed v_s (speed of sound). A single hit creates a single shock wave that will be perceived as a short click. To create a continuous sound like voice, the body should vibrate aka repeatedly hit medium for the duration of sound (Figure 3.1.a). The rate at which a body hits the medium determines a pitch of a sound.

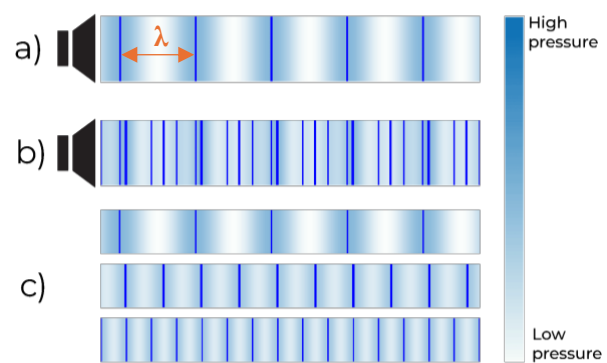


Figure 3.1 – Different types of pressure waves that can propagate in a medium. a) pressure wave created by a speaker that continuously hits a medium with constant period λ . Blue lines mark the spot where the hit occurred. b) a complex pattern with which speaker hits a medium. c) a set of simple periodic hit patterns that make up the complex pattern in b). Notice that all the blue lines in b) have a corresponding blue line in c).

The simplest pattern (oscillation) that has a constant pitch is when hits appear at a constant rate. More complex patterns are certainly possible, like one in Figure 3.1.b. Interestingly, one can notice that there are 3 elementary oscillations that make up that complex pattern (Figure 3.1.c). In mathematics, representing a signal as a sum of elementary oscillations is called Fourier transform. Different waves will have different sets of oscillations, but any wave can be represented this way. Thus, one might say that sound on the Figure 3.1.b has several pitches sounding at once.

From now on we will refer to such elementary periodic oscillations like on Figure 3.1.c as **partials** of a sound. Each partial can be characterized by a number λ – wavelength of oscillation. Wavelength is a distance in meters in which waveshape repeats (Figure 3.1.a). Knowing a speed of sound in a medium v_s and a wavelength λ one can find a frequency of oscillation f using the following formula:

$$f = \frac{v_s}{\lambda} \quad 3.1$$

Frequency represents a speed of oscillation or how many hits are happening per second. The inverse relationship between λ and f means that low frequency instruments, like bass guitar, produce sounds with partials of longer wavelength than high frequency instruments like flute. Because each medium has unique speed of sound, the same body vibrating with the same frequency will generate waves of different wavelengths in different mediums. Thus, frequency is a better characteristic of a vibration as it is medium independent.

The remarkable thing about human biology is that part of inner ear called cochlea performs Fourier transform of the sounds that enter it before sending information to the brain [4]. So rather than hearing a complex sound like on Figure 3.1.b, the brain works with a set of partials like in the Figure 3.1.c. Moreover, most musical instruments (including our voices) create complex vibrations containing dozens of partials even when a single pitch is being played. This gives us a clue that using a single number to represent a musical pitch may lack important information.

3.2 Sound as a set of frequencies

One of the most fundamental concepts in mathematics is a concept of set. Informally, one can define a set as an unordered collection of unique mathematical objects, be it numbers, vectors, other sets etc. Sets are denoted as capital letters and elements of the set are specified in curly brackets like so: $S = \{ 2, 7, 6, 1.33 \}$. Here S is a set of numbers 2, 7, 6, 1.33. Another way to specify a set is by stating properties that define its elements: $S = \{ s \mid s \in \mathbb{N}, s < 5 \}$. This is read like this “The set S is a set of all s such that s is a natural number and is less than 5.”, which means that $S = \{ 1, 2, 3, 4 \}$. Refer to the Table 8.1 and Table 8.2 in supplementary materials for the list of symbols and operations on sets used in this article. Operations in the Table 8.2 are specific to this paper and are not used in the common set theory.

As we previously discussed, real musical instruments produce complex vibrations that consist of many partials each with unique frequency. Thus, rather than using a single number to represent a pitch of a sound we can represent it as a set of frequencies of all its partials:

$$F = \{f_1, f_2, \dots, f_k\} = \{f_i \mid i = 1, 2, \dots, k\}$$

3.2

Here, F is a set that represents a sound that is made of k frequencies f_i . The example of what that means is shown in Figure 3.2. The complex wave in Figure 3.2.a shows how the pressure on Y-axis varies with time on X-axes, that is called **time-domain** representation. It can be transformed into **frequency-domain** representation using Fourier transform (black line in Figure 3.2.b). The amplitudes and frequencies of individual partials that make up the sound are marked by the peaks on the Fourier transform graph. However, there is always implicit uncertainty in Fourier transform derivation of partial frequencies and their amplitudes. The precision with which

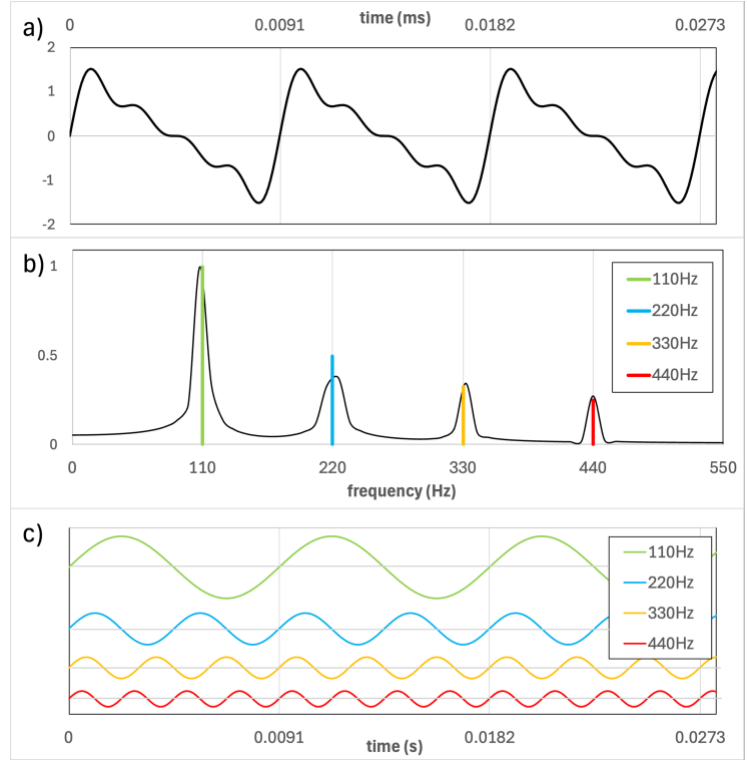


Figure 3.2 – Demonstration of how a complex wave can be seen as a collection of elementary oscillations (partials) in both time and frequency domains. a) The time-domain representation of a complex wave. b) the black line is Fourier transform or a frequency-domain representation of the wave in a). Colored vertical lines represent frequencies and amplitudes of partials that comprise the wave in a). c) The time-domain representation of individual partials that comprise the wave in a), colors correspond to b).

those amplitudes and frequencies are derived depends on the input data. The exact frequencies that build up the sound in Figure 3.2.a were known in advance and are shown as colored vertical lines in Figure 3.2.b. Such vertical lines is how we will be presenting the frequencies and amplitudes of partials in frequency domain. The time domain representation of partials in Figure 3.2.b is shown in Figure 3.2.c Summing up all the graphs in Figure 3.2.c will yield exactly the graph in Figure 3.2.a.

As we saw, every partial in a sound is characterized by 2 numbers, frequency and amplitude. However, for simplicity, we will not include amplitude into consideration and limit our analysis to frequencies only. Thus, the set of frequencies that corresponds to the sound in Figure 3.2 is following:

$$F = \{110, 220, 330, 440\} \quad 3.3$$

3.3 Harmonic sets

There are as many different sets of frequencies as there are sounds. Some sets may appear structured, others may seem random. Majority of musical instruments produce **harmonic** sounds in which each frequency is an integer multiple of a **fundamental** frequency. The sound in Figure 3.2 is an example of such sound because every frequency in the Formula 3.3 is an integer multiple of 110 Hz , which is a fundamental frequency of that sound. From now on, we will denote fundamental frequency as f_F . Because harmonic sounds are very common in music and follow a particular structure, we can use definitions from Table 8.2 to rewrite the set F more compactly $F = 110 \cdot \{1, 2, 3, 4\} = 110 \cdot \mathbb{N}_4$. In general, we can define harmonic sounds as such that can be described by the following formula:

$$F = f_F \mathbb{N}_k \quad 3.4$$

One of the features of harmonic sounds is that they are perceived as a single pitch with the frequency of f_F . The number of frequencies k determines the timbre of the sound and depends on many factors such as the intensity with which the note is stricken. The larger the number k the richer and brighter the timbre is. In reality, the number k changes with time as high frequencies tend to decay faster than low frequencies. We will not pay attention to that dynamism and suppose that a typical note played on a musical instrument has a fixed fundamental frequency and a fixed small number of partials k .

It is clear from the Formula 3.4 that fundamental frequency is an elementary vibration from which all other frequencies in a set are made of. In mathematics, the number that serves that purpose is known as **greatest common divisor** (**gcd**), thus it can be written that:

$$f_F = \text{gcd}(F) \quad 3.5$$

The geometric meaning of **gcd** is shown in Figure 3.3. One can see that **gcd** is sort of a building block for a set of numbers.

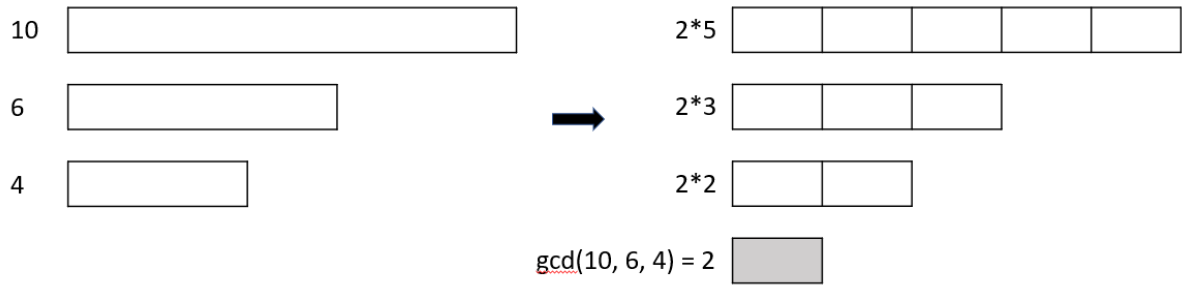


Figure 3.3 – Geometric meaning of the greatest common divisor (gcd). On the left there are 3 numbers 10, 6 and 4 represented as units of length. On the right same numbers are the results of stacking the $\text{gcd}(10, 6, 4) = 2$ a whole number of times. It is evident that gcd is a number that can be viewed as a unit length from which other numbers can be made. Greatest common divisor does not have to be a whole number.

The key feature of the harmonic sounds is periodicity. Example of that can be seen in Figure 3.2.a where the waveform repeats with a period of $T_{total} = 9.1 \text{ ms}$. The reason for that periodicity is purely geometrical. At some point the periods of all constituting partials coincide, that resets the wave to its initial state and causes the pattern to repeat. That mechanism is seen from comparing periods of oscillations in Figure 3.2.a and Figure 3.2.c.

In mathematics, the smallest number to which some other numbers fit whole number of times is called **least common multiple** (lcm). It can be used to find the period of the total wave from the set of periods of its partials. To do it we need to convert a set of frequencies to a set of periods:

$$T(F) = \left\{ \frac{1}{f_i} \mid f_i \in F \right\} \quad 3.6$$

and find the lcm

$$T_{total} = \text{lcm}(T(F)) \quad 3.7$$

Let's notice the period of a total wave in Figure 3.2.a is the same as the period of fundamental in Figure 3.2.c. That is a consequence of a reciprocal nature of frequency and period, and of the geometric meaning of **gcd**. Let's look at the example of a harmonic set $F = \{f_F, 2f_F, 3f_F, \dots, kf_F\}$. The set of periods for F according to the Formula 3.6 will be $T(F) = \left\{ \frac{1}{f_F}, \frac{1}{2f_F}, \frac{1}{3f_F}, \dots, \frac{1}{kf_F} \right\}$. We can substitute the period of fundamental $T_F = \frac{1}{f_F}$ and write $T(F) = \left\{ T_F, \frac{T_F}{2}, \frac{T_F}{3}, \dots, \frac{T_F}{k} \right\}$. Because each consecutive period is a quotient of T_F on a natural number, the

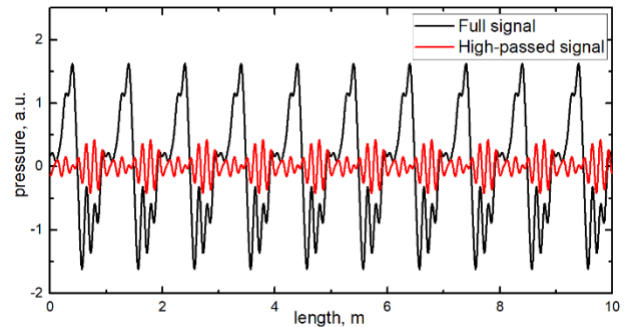
$\text{lcm}(T(F))$ is the T_F . On the other hand, $T_F = \frac{1}{f_F}$. Thus, the total period of the wave can be found using **gcd** instead of **lcm**:

$$T_{total} = \frac{1}{\text{gcd}(F)} \quad 3.8$$

This way the role of fundamental is twofold. On one hand, the frequency of fundamental is the building block for the rest of the frequencies in a set. On the other hand, it has the same period as the total wave. That means that if different sets have the same fundamental, they will have the same period and vice versa.

3.4 Phantom fundamental, inharmonicity and periodicity

Not every sound has a real fundamental. Let's take a set $F = \{440, 550\}$. Using the Formula 3.5 we get $f_F = \text{gcd}(\{440, 550\}) = 110$. But 110 Hz is not the element of set F , or $f_F \notin F$. In such case it is said that the sound has **phantom fundamental** of 110 Hz. The phantom fundamental has the same properties as real fundamental. It is also the building block of the rest of the frequencies in the set and it has the same period as the total wave (Figure 3.4).



The surprising thing is that perception of the pitch of the total wave does not change if the sound has real or phantom fundamental. Meaning that sounds in Figure 3.4 are perceived to have the same pitch even though the phantom fundamental is not directly present in the sound.

Figure 3.4 – The black line is the waveform of the harmonic spectrum with 7 partials including fundamental frequency. Red line is a waveform of the same spectrum but with first 4 partials removed. Note that the periods of two waves are the same. The period of the black line equals to the period if its fundamental and the period of red line to its phantom fundamental which are equal in this example.

Strictly speaking, for some sets the **gcd** may not exist. If a ratio between any two frequencies in a set is irrational, no matter how small the building brick (**gcd**) we pick, we won't be able to build up both of those frequencies with it. In mathematics, the numbers a and b are called **commensurable** if their ratio a/b is rational. Geometrically that means that for such numbers there is a common unit of length c that can be used to measure both a and b , $a = nc$ and $b =$

mc where $n, m \in \mathbb{Z}$. That is necessary and sufficient condition for the existence of $\gcd(a, b)$. If a/b is irrational, numbers a and b are called **incommensurable** and $\gcd(a, b)$ does not exist. Thus, we can say that if the set F is commensurable, meaning all frequencies in F are commensurable, there exists $\gcd(F)$.

If for a set F there exists no $\gcd(F)$, such sound does not have a fundamental and the wave of such sound is not periodic. While such case is possible in mathematics, it is not possible in practice because achieving irrational ratio requires an infinite precision. Nevertheless, some combinations of partials can produce phantom fundamentals with wavelengths many thousands of times larger than that of constituting partials. That pushes fundamental outside of a hearing range and makes the sound to be perceived as non-periodic.

The Figure 3.5 demonstrates how period can change with the structure of a set. Sounds in Figure 3.5.a and Figure 3.5.b have a simpler ratios between partials, which results in smooth and stable sound and high periodicity. Seemingly similar set in Figure 3.5.c has much longer period and thus will be perceived as unstable sound with noticeable variations in volume.

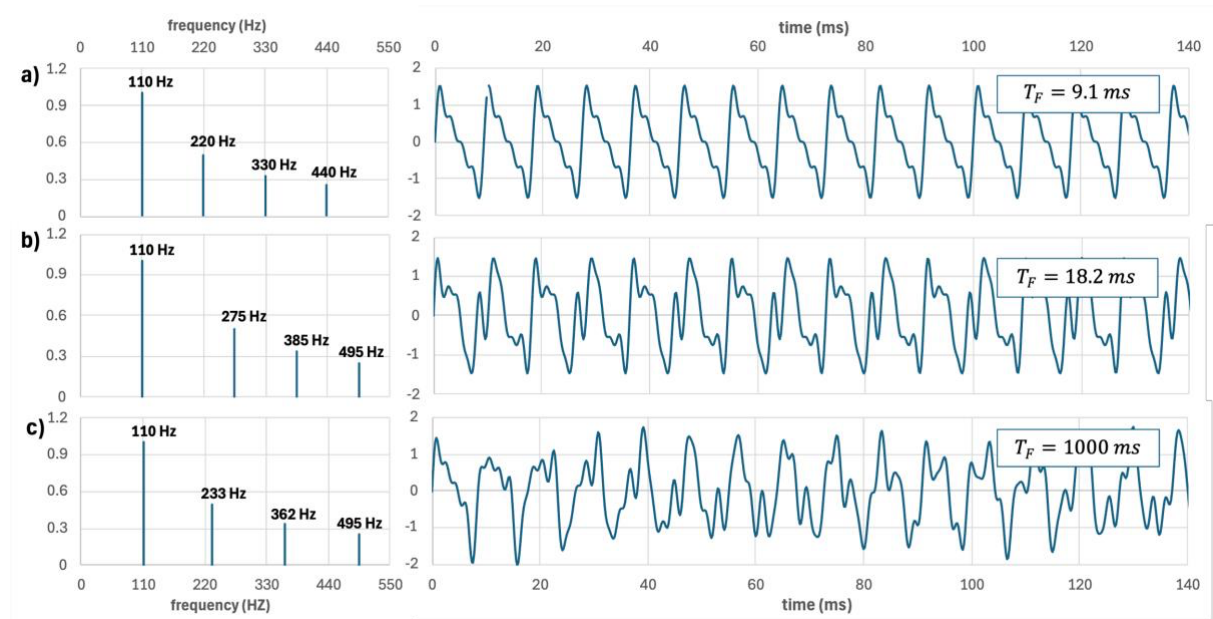


Figure 3.5 – The relationship between set of frequencies and the period of a sound wave. a) harmonic spectrum with fundamental $f_F = 110 \text{ Hz}$ with a period of $T_F = 9.1 \text{ ms}$. b) inharmonic sound with second, third and fourth partials shifted to higher frequencies. That shift results in emergence of phantom fundamental $f_F = 55 \text{ Hz}$ with period $T_F = 18.2 \text{ ms}$. c) inharmonic sound that may seem more uniform in frequency domain than the sound in b) but with significantly lower phantom fundamental that is outside of a hearing range $f_F = 1 \text{ Hz}$ with a period of $T_F = 1 \text{ s}$. Such large period results in the perception of instability of overall sound.

The sounds that deviate from harmonic set such as in Figure 3.5.b and Figure 3.5.c are called **inharmonic** sounds. Britannica defines inharmonicity as “deviation of the frequencies of the

harmonics from the exact multiples of the fundamental”. That assumes a different definition of fundamental than the **gcd** of a set of frequencies, as in the case of **gcd**, frequencies of the harmonics cannot deviate from integer multiples by definition. What is called a fundamental in modern empirical tradition is the most pronounced low-frequency partial of a sound that is perceived as a pitch of that sound. From that perspective, all sounds in Figure 3.5 have the same fundamental of 110 Hz, because they will be perceived as a same pitch but with a different timbre.

Which definition of fundamental to pick for a theory depends on the philosophical tradition that you submit to. I believe that strict mathematical definition of fundamental reveals a geometrical meaning behind relationship of frequencies within a sound. Some frequencies fit together forming periodic patterns while others don’t. That will help us mathematically define consonance in Chapter 4.3 and use that knowledge to build a tuning theory. The fact that perceptions may deviate from pure mathematics should be accounted separately by modelling perception on the basis of the idealistic framework. Building such model is not the task of the current paper.

3.5 Defining notes as sets

As was mentioned in Chapter 2.3, the new tuning framework has to be as convenient as possible for modern musician to have a chance for success. Thus, it is important to have a system of pitches that is familiar but at the same time more flexible than the common one. Modern 12-tet tuning was made for keyboard instruments and just like piano keyboard has a total of 12 notes that repeat for every octave. Each note has a name (A4, C#5, etc.) and the corresponding frequency that form a very strict grid.²⁸ In reality, acoustical musical instruments, including pianos, cannot be tuned perfectly to that grid, so some pitch deviations are inevitable. If a deviation of note from that grid is big enough to be picked up by a human ear, the note is considered to be out of tune. For example, the frequency for C#5 on a piano should be 554.37Hz. If we tune that note to 550Hz instead, which is a perceivable difference, the instrument will be considered out of tune.

Such tuning of instruments to a predefined grid of pitches is a reality of tuning to most modern musicians. Moreover, modern tuning practice does not assume that there are other notes outside of that grid. If you follow it – you are in tune, if you don’t – you are out of tune. That is a reality that we have to face, and in order to make a system more accessible to modern musician I suggest

²⁸ The grid of 12-tet pitches has the note A4 tuned to exactly 440Hz and all other notes are determined by shifting that frequency the correct number of semitones away. For example, C#5 is 4 semitones higher than A4, thus it will have a frequency $440 \cdot 2^{\frac{5}{12}} \approx 554.37 \text{ Hz}$.

sticking to established 12 notes notation but easing out the strictness of frequency correspondence. We will assume that a note corresponds to a range of frequencies, rather than a specific frequency. A range for a note is calculated as ± 50 cent difference from pitches of 12-tet grid. This way, if a pitch is within 50 cents from 554.37Hz, it will be considered C#5. The full list of frequency ranges is presented in the Table 8.3.

Having stated that, we can come up with a way of marking which note a given set corresponds to. Let's consider the sound in Figure 3.2 as an example. The fundamental frequency of that sound is 110 Hz , which according to Table 8.3. corresponds to the note A2. We can use that note as a name for the set instead of generic F . We can also use a subscript index to specify the number of partials k that note has. This way, the set F in the Formula 3.3 can be rewritten as $A2_4$, which clearly indicates that F is the A2 note made of 4 harmonic partials.

In 12-tet system defining frequency of 1 note, defines frequencies of all notes as intervals between them are fixed. In our system we have to define other notes by a relative interval from a note with a known frequency. This way, if $A2_4 = 110 \cdot N_4$, we can define $E3_4 = \frac{3}{2}A2_4$, which is just fifth up from $A2_4$, or a $C\#3_4 = \frac{5}{4}A2_4$, that is just major third up from it.²⁹ That is a very convenient way of notation within set theoretic approach that will be used in following chapters. It also allows for a clear notation of sets that correspond to chords. For example, the pitches we defined previously form a just major triad. The set for that triad is a union of sets of all notes that form it, or $F_{triad} = A2_4 \cup C\#3_4 \cup E3_4$.

Set theory seems like a suitable language to describe a tuning theory with. The most important take away from that chapter is that a pitch is better represented as a set of numbers rather than a single number. That realisation is a key to mathematical definition of the phenomenon of consonance that will be proposed in the next chapter.

²⁹ Notice, that once we defined $E3_4$ we can define $C\#3_4$ relative to it, instead of $A2_4$. $C\#3_4$ is a just minor third down from $E3_4$ or $C\#3_4 = \frac{5}{6}E3_4$, which is exactly the same as $C\#3_4 = \frac{5}{4}A2_4$. There is no correct way of doing such definitions, it depends on the context of what you are trying to achieve.

4 Measures of consonance

In this chapter I will suggest two measures of consonance for arbitrary sets of frequencies that represent sounds. Those measures are harmonicity and affinity. Affinity measures how closely related different sounds are, and consequently how well they fit to each other when sounding simultaneously. Harmonicity measures to which extent a set of frequencies forms a single pitch. Both measures are numbers between 0 and 1, where 1 means perfect consonance and 0 absolute lack of consonance. I will also compare those two measures with the similar theories from literature.

4.1 The definition of consonance

To build mathematical measures of consonance let's recall the definition of consonance given in the Chapter 2.1. We can restate it in a following way.

Consonance is an observation that some sounds when played together sound:

1. smooth and stable,
2. harmonious as if they fit together forming one larger whole.

Here we divide the definition into two parts. The first part describes sensory aspect of consonance, an observation of smoothness and stability. The second part relates to the philosophical problem of unity and multiplicity, the mysterious phenomenon when many become one.

Within idealistic framework the second aspect of consonance must be held as primary from which the first, sensory, aspect should unfold. Thus, let's state the definition of consonance for which we will seek a mathematical description. I suggest it should sound like this:

**“Consonance - is a phenomenon when multiple sounds fit together
forming one greater harmonious whole.”**

A mathematical expression of that idea can serve as a measure of consonance. Successful measure, when applied in practice, should produce sounds that are perceived as smooth and stable. Many such measures can be suggested. We should prefer the simplest ones that better describe the empirical data.

4.2 The affinity of sets

In the Chapter 3 we concluded that appropriate mathematical object to describe a sound is a set of frequencies. One of the aspects of our definition of consonance states that consonant sets are the ones that fit together. We can say that sets fit together if they share something in common with each other. Mathematically that commonality can be measured as the relative size of a shared fraction between different sets. If the proportion of shared elements is big - then sounds are closely related and if it is small, then not so much. We will call that measure of shared elements as **affinity**³⁰, and it will serve as a first measure of consonance. The simple way to quantify affinity is the following formula:

$$\alpha(F, F') = \frac{|F \cap F'|}{\min(|F|, |F'|)} \quad 4.1$$

Here $\alpha(F, F')$ is the affinity of sets F and F' . Both of those sets are completely arbitrary and thus can represent any sound, harmonic or inharmonic, a single note or a complex chord. In the numerator we have the size of the intersection of F and F' , or the number of common frequencies. If there are no common frequencies between F and F' , then the intersection is an empty set $F \cap F' = \emptyset$. The size of the empty set is zero $|F \cap F'| = 0$, which leads to zero value for the affinity $\alpha(F, F') = 0$. That case signifies absolute lack of consonance between F and F' . Graphically that case is represented in the Figure 4.1.a.

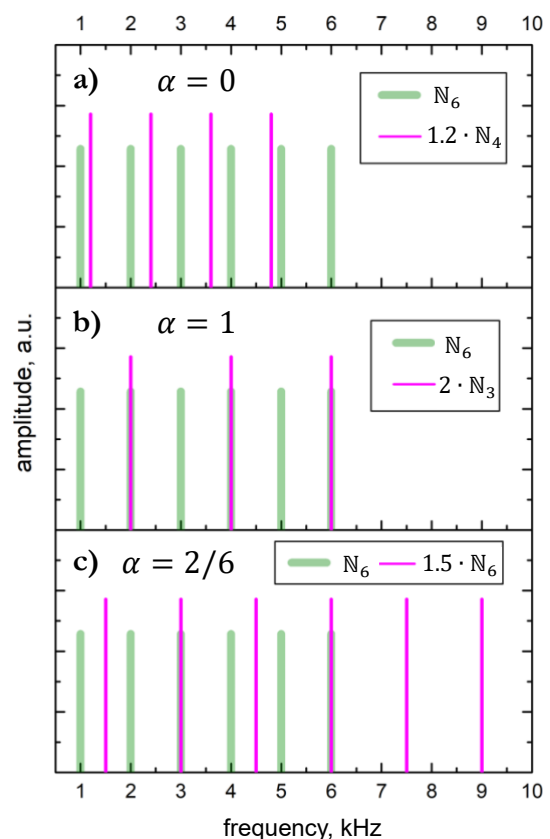


Figure 4.1 – Demonstration of how different sets can have different values of affinity. a) Two sets N_6 and $1.2 \cdot N_4$ do not share any frequencies, thus having a 0 value of the affinity. b) A set $2 \cdot N_3$ is a subset of N_6 , thus the affinity is equal to 1. c) The realistic case of perfect fifth, two sets N_6 and $1.5 \cdot N_6$ have 2 partials in common, which makes the affinity to be $2/6$.

If there are some common elements, then affinity has a positive value. To determine it, we have to calculate the denominator which is the size of the smallest of two sets F and F' . It makes sense

³⁰ Affinity as a measure of consonance should not be confused with the notion of affinity in music theory that refers to scales and scale degrees in tonal music. Those are different notions that may or may not have a deeper connection between them.

to relate the size of the intersection to the smallest set, as in this way we will get 1 as a maximum possible value for affinity. That case is demonstrated in Figure 4.1.b when the smaller set is a subset of a larger one. Let's see how we get the value for affinity in that case. If $F' \subseteq F$, then in the numerator the size of the intersection will be the size of the subset $|F \cap F'| = |F'|$. As for denominator, the size of a subset is always less or equal to its superset, thus $\min(|F|, |F'|) = |F'|$. Consequently, the value of affinity becomes $\alpha(F, F') = \frac{|F'|}{|F'|} = 1$, which means the perfect consonance between two sets.

As we see in later analysis, in practice most musical intervals have an intermediary value between 0 and 1 such as shown in Figure 4.1.c. That value depends on the number of elements in the smallest set and the size of the intersection. Affinity is a very simple measure of consonance that gives us a numeric value by which we can check consonance of different combinations of sounds and build a tuning system with it.

Affinity is very similar to the idea of spectral interference [4], [2], though it has some important differences. Sethares's work is based on the limited resolution of human auditory system that was empirically determined in [5]. Using results of that work, Sethares proposes a way to calculate a dissonance value for arbitrary set of frequencies. If the ear can resolve all frequencies in a set, then the dissonance is low. If there are a lot of frequencies that cannot be resolved, then the dissonance is high. Sethares applies that idea to tuning theory by calculating which intervals have relatively low values of dissonance. That is done by drawing dissonance curves. To draw them, Sethares takes a static sound A (a note or a chord) and some other sound B that we might want to add to A . Then he sweeps a range of intervals t and checks the dissonance value of combination of A and tB for

each value of t . The result of such sweep looks like a curve with maximums and minimums like one in Figure 4.2. Sethares proposes that a good interval to use in tuning is the one with relatively low dissonance value. In other words, a local minimum of dissonance. He notices that there are two types of minima: sharp minima at which partials of A and tB coincide, and broad minima where partials of A and tB are sparse and easily resolved.

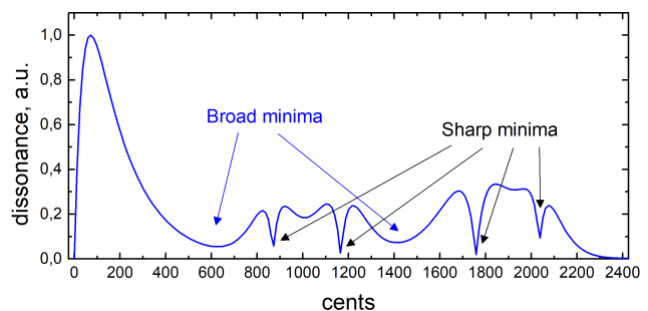


Figure 4.2 – Dissonance curve of an inharmonic sound that demonstrates two types of dissonance minima: sharp minima, where partials of two sounds coincide and a broad minima, where dissonance is low due to large distance between partials, such that the ear is able to resolve them as separate.

The dissonance in Sethares's work was defined as a rough sound due to high degree of beating between partials. Thus, the intervals that corresponded to the minima of dissonance were characterized by a relatively smooth sound. Sharp minima on a dissonance curve are caused by coincidence of partials of two sounds, which is exactly the condition for non-zero affinity. As we will see in the Chapter 5, tuning derived from dissonance curves can be identical to the one derived by using affinity as measure of consonance. Thus, we should expect sounds with high affinity to also be relatively smooth sounding, which satisfies our expectation for the correct consonance measure.

Even though spectral interference and affinity approaches have similarities, they are radically different in their philosophical foundation. Sethares theory is empirical theory and affinity-based theory is idealistic. One of the upsides of the affinity-based approach is dramatic reduction of computation needed to derive tuning. But most importantly, idealistic theory is biology independent.

In his work Sethares argued that capturing broad minima of dissonance is the important feature of dissonance curves. However, I believe that those broad minima introduce subjective element to his theory. Let's imagine that due to some sickness, the frequency resolution of inner ear decreases. This means that parameters that describe that resolution in dissonance curve derivation should also change. As seen in Figure 4.3, the decrease of resolution can lead to disappearance of broad minima (orange curve) or even to flattening of sharp minima (red curve). That raises the question, if one becomes deaf does musical tuning disappear? In idealistic theory, no matter what frequencies are sounding or who is listening, tuning is the same and is determined by the relationship between mathematical objects.

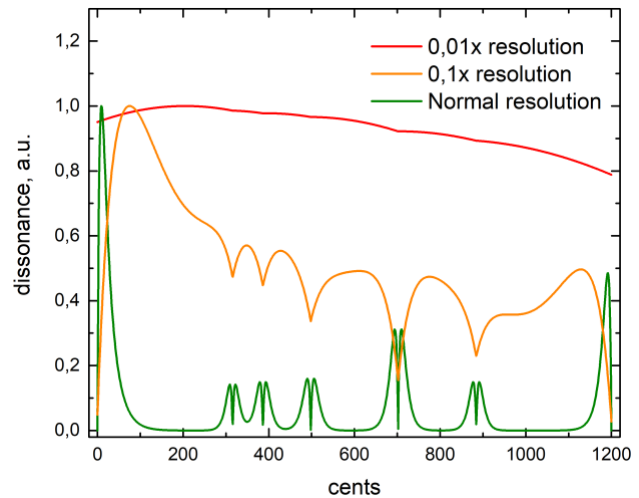


Figure 4.3 – Dissonance curves with parameter χ^* varied to demonstrate the lowering of resolution of auditory system. The green curve represents normal frequency resolution with point of maximum dissonance $\chi^* = 0.24$ as used in [4]. Lower resolution curves are obtained by lowering that value to 0.03 for the orange curve and 0.003 for the red curve. As resolution drops, first broad minima disappear, and then sharp minima become much less pronounced.

4.3 The harmonicity of sets

The affinity contribution alone does not satisfy the definition of consonance given in the beginning of this chapter. It accounts for sets fitting together, but not for the formation of a greater harmonious whole. To account for that, let's notice that each note is perceived by us as a single pitch even though it is made of many pitches. As we discussed in Chapter 3.3, there is a special arrangement of pitches that is perceived by us as a single pitch. That arrangement is a harmonic set that is represented by the Formula 3.4.

As we discussed before, such sets have a lot of interesting properties and can serve as a great abstraction to represent notes played on real instruments. The pattern of integer multiples of fundamental frequency is commonly known as harmonic series and is used as a basis of many tuning theories, most notably of JI. It is our assumption that is backed by common experience that a set like in Formula 3.4 represents a single pitch. What is interesting, is that any arbitrary set can be compared to how well it resembles harmonic set. We will call the measure of such resemblance

harmonicity.³¹ The larger the resemblance of a set of pitches to the harmonic set - the greater is the oneness of total sound and therefore, the better they form one greater harmonious whole.

In order to find harmonicity of a set, we need a harmonic set to which we can compare a given set to. Let's remember that for any commensurable set of frequencies F we can find a fundamental frequency as $f_F = \gcd(F)$. As we discussed in the Chapter 3.3, all frequencies in F are integer multiples of f_F . Thus, we can say that there exists a set $N \subseteq \mathbb{N}$ such that $F = f_F N$. If F is finite, we can find a number $n = \frac{\max(F)}{\gcd(F)}$ that is large enough so that $N \subseteq \mathbb{N}_n$. Knowing that and using the Formula 3.4 we can find a superset of F like so:

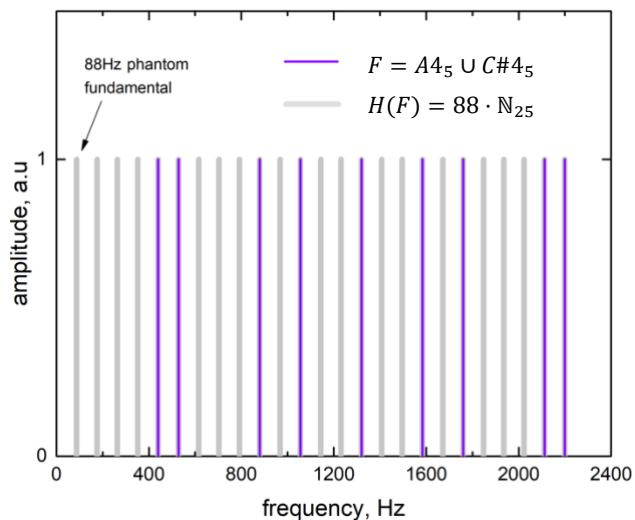


Figure 4.4 – Demonstration of harmonicity as a measure of consonance for the interval of major third. The violet lines represent a spectrum of a major third interval that is made as a union of the note $A4_5 = 440 \cdot \mathbb{N}_5$ and $C\#5_4 = 5/4 \cdot A4_5$ with frequencies $F = A4_5 \cup C\#5_4$. The grey lines represent the corresponding harmonic superset $H(F) = F2_{25} = 88 \cdot \mathbb{N}_{25}$. One can see that F is the subset of $H(F)$ with the harmonicity value of $\chi(F) = 9/25$. That means that frequencies that comprise a major third interval indicate towards a single pitch with frequency of 88Hz but do not form it, they are only a part of a larger whole.

³¹ The similar metrics in literature have different names such as tonalness [4] or harmonic similarity [7]. It can be categorized as pattern-matching harmonicity/periodicity theory according to [1].

$$H(F) = f_F \mathbb{N}_n, \quad n = \frac{\max(F)}{\gcd(F)} \quad 4.2$$

The set $H(F)$ is a harmonic set, which means it is a single pitch and F is a subset of $H(F)$. Thus, any set F with commensurable frequencies is a subset (a part) of a single pitch $H(F)$. We will call $H(F)$ a **harmonic superset** of F . The meaning behind harmonic superset is illustrated in Figure 4.4, where a set of the just major 3rd made of notes $A4_5 = 440 \cdot \mathbb{N}_5$ and $C\#4_5 = 5/4 \cdot A4_5$, is a subset of a single lower pitch $F2_{25} = 88 \cdot \mathbb{N}_{25}$ with 25 partials. We can use the relative size of a set F to its harmonic superset $H(F)$ as a measure of harmonicity:

$$\chi(F) = \frac{|F|}{|H(F)|} \quad 4.3$$

Strictly speaking, not every set has a corresponding harmonic superset. If F is incommensurable there is no $\gcd(F)$ and thus, the $\chi(F)$ is undefined. We can treat it as a special case and assume harmonicity of 0 for all incommensurable sets. The Formula 4.3 can be rewritten by replacing $|H(F)|$ with $\frac{\max(F)}{\gcd(F)}$ according to Formula 4.2 and writing $\chi(F)$ so that there are two sets as inputs, like in the Formula 4.1 for the affinity. This way we get the following formula for harmonicity:

$$\chi(F, F') = \begin{cases} \gcd(F \cup F') \frac{|F \cup F'|}{\max(F \cup F')}, & \exists \gcd(F \cup F') \\ 0, & \nexists \gcd(F \cup F') \end{cases} \quad 4.4$$

This formula accounts for both commensurable and incommensurable sets, which is relevant for mathematical strictness but is not relevant for real world sounds, as any physical frequency set is commensurable. Just like affinity α , harmonicity χ is a number between 0 and 1, where 0 means absolute lack of harmonicity and 1 means that F and F' form a single pitch.

The idea behind the harmonicity measure χ may be categorized as pattern-matching theory according to [1]. It is identical to the measure of harmonicity from the works [6] and [7] but reformulated to set theoretic language. The work [6] focuses on rank ordering scales by their harmonicity and the work [7] ranks chords and intervals by the same measure. Both studies show that harmonicity defined in such a way serves as a great predictor of consonance preferences. In

this work I view harmonicity as an internal implicit characteristic of an arbitrary set of frequencies. It can be applied to inharmonic sounds, intervals or chords and can be used to determine dynamic tuning on the fly, rather than ranking static tunings. The idea behind that measure is that individual pitches in the set may or may not form a larger single pitch, which is defined as harmonic set. If individual pitches relate as irrational numbers, they are completely disjoint and do not take part in larger whole. If relationship is rational then they either form a single pitch or a subset of it. The value of harmonicity χ measures how close a set of pitches is to the single pitch.

Sounds with high degree of harmonicity are perceived smoother and more stable, which satisfies our expectation for correct measure of consonance. The reason for that is seen in Figure 3.5, as harmonic sounds have more regular and periodic waveforms than other sounds.

One aspect that can be improved is the definition of what sets constitute a single pitch. We only consider sets that satisfy the Formula 3.4 as being single pitches. Thus, removing one or several arbitrary partials from $f_F \mathbb{N}_k$ will lower the harmonicity. That makes such common timbres as a square wave or high-passed sounds to have an intrinsic degree of dissonance. That is counter intuitive and perhaps is too strict. However, I do not have a better suggestion for this measure of consonance than the one defined by the Formula 4.4.

There may be many ways to quantify consonance in terms of set theoretic approach. The ones used in the article are the simplest ones that are closely related to the prominent empirical theories. I admit that there is plenty of room for improvement as many properties of elementary oscillations were omitted, such as amplitude and phase information. There may be a big impact on consonance metrics from those additional parameters. I welcome anyone interested to participate in improving the theory. My goal is to have a simple, practically applicable theory that can drive new instrument and music production.

5 From consonance to tuning

In the Chapter 1.2 we set out to find a natural solution to the tuning problem. That solution is a tuning system that arises naturally from mathematical objects and is independent from human values or perceptions, while at the same time being in harmony with them. We postulated a definition of consonance and suggested two quantities: affinity α and harmonicity χ that could serve as measures of consonance. Using them allows us to determine a set of intervals, at which any two arbitrary sounds harmonize with each other, thus giving us a natural tuning system. Individually, affinity and harmonicity generate different sets of intervals, thus providing very rich framework for tuning derivation. That flexibility can serve as an artistic tool to find new ways of expression. In following chapters I will suggest the simplest ways these consonance measures can be used to derive tuning systems and discuss their advantages and disadvantages.

5.1 Affinitive tuning system

At first let's see what intervals are generated from the affinity. In the Formula 4.1, affinity is a function of 2 arguments, which are sets F and F' . Let's refer to F as **contextual set**. It will represent frequencies of a note or a chord that serves as a musical context. The F' is a **complementary set** to that context. So, let's think of F' as sounding relative to F , in context of F . As a real-life example, we can think of a choir where F is a drone note that creates context for a melody. A note in a melody is represented by F' . Without context, we cannot speak about tuning because any pitch is allowed. Tuning is a relative phenomenon, not an absolute one.

Both F and F' are arbitrary sounds. If it happens that they share some frequencies, they will have a non-zero value of affinity $\alpha(F, F')$ according to Formula 4.1. However, if they don't, we can always transpose F' on some interval t so that $\alpha(F, tF') > 0$. Let's think of a choir example again. We have a drone note F relative to which another singer is about to take a note F' . At first the singer misses the note, so that $\alpha(F, F') = 0$, which sounds out of tune. But quickly he or she corrects it by shifting F' on an interval t , so that frequencies of both notes align and $\alpha(F, tF') > 0$. Depending on the structure of F and F' there can be many such intervals with non-zero affinity. The set of all such intervals we can call affinitive intervals $\mathcal{A}(F, F')$ and define as:

$$\mathcal{A}(F, F') = \{t \mid \alpha(F, tF') > 0\} \quad 5.1$$

Both F' and F are arbitrary sets that can represent a single note or a chord, be harmonic or inharmonic. The difference is that set F represents a static, contextual sound; and F' is the sound that we want to add relative to F . The set of affinitive intervals $\mathcal{A}(F, F')$ is all the intervals we can transpose F' on so it fits to F .

It may seem that finding $\mathcal{A}(F, F')$ is a difficult task but it's actually quite trivial. Let's remember that according to Formula 4.1 affinity of two sets is 0 when their intersection is an empty set. Thus, we can rewrite 5.1 to be:

$$\mathcal{A}(F, F') = \{t \mid F \cap tF' \neq \emptyset\} \quad 5.2$$

It is evident, that in order to get non-empty intersection, a number t must be such, that upon multiplication on an element of set F' it should produce an element of F , or $tf' = f$ where $f \in F$ and $f' \in F'$. Solving this simple equation for t gives us the formula for the set of affinitive intervals:

$$\mathcal{A}(F, F') = \left\{ \frac{f}{f'} \mid f \in F, f' \in F' \right\} \quad 5.3$$

In simple words, a set of affinitive intervals is a set of ratios of every frequency in the contextual set F on every frequency of the complementary set F' .

Let's see how that can be applied in practice. For simplicity, let's assume that both sets are simple harmonic sets with 6 partials that represent the note $C4$ with fundamental of 100Hz.

$$F = F' = C4_6 = 262 \cdot \mathbb{N}_6 \quad 5.4$$

According to Formula 5.3 affinitive intervals are the ratios between every element of F and F' . The easy way to find them is to build a table, like the Table 5.1, where each cell is a simplified ratio of an element of F (top row) to an element F' (first column).

Table 5.1 - Derivation of affinitive intervals for the sets F and F' defined by Formula 5.4.

$F \text{ (Hz)} \backslash F' \text{ (Hz)}$	262	524	786	1048	1310	1572
262	$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$	$\frac{5}{1}$	$\frac{6}{1}$
524	$\frac{1}{2}$	$\frac{1}{1}$	$\frac{3}{2}$	$\frac{2}{1}$	$\frac{5}{2}$	$\frac{3}{1}$
786	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{1}$	$\frac{4}{3}$	$\frac{5}{3}$	$\frac{2}{1}$
1048	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{1}$	$\frac{5}{4}$	$\frac{3}{2}$
1310	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{1}{1}$	$\frac{6}{5}$
1572	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	$\frac{1}{1}$

As you can see from the table, there are in total 36 intervals but many of them repeat more than once. Because sets do not contain duplicate elements, the actual number of unique elements is 23 and the set looks like this:

$$\mathcal{A}(C4_6, C4_6) = \left\{ \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, 1, \frac{6}{5}, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, 2, \frac{5}{2}, 3, 4, 5, 6 \right\} \quad 5.5$$

This is a complete list of intervals for which $C4_6$ will harmonize with $C4_6$. The frequency structure of the note naturally elevates just 23 numbers from the infinite list of possibilities. Changing the contextual or complementary sets will elevate other intervals in a mathematically strict manner.

Traditionally, having a set of affinitive intervals would be considered as having an affinitive tuning system, but set theoretic approach can be pushed further. Each interval can be paired with a corresponding value of consonance. This way the tuning system becomes a hierarchical system of intervals, from most to least consonant. In addition to calculating a value for affinity for each interval, nothing stops us to account for harmonicity as well. We can do it by finding an arithmetic mean of affinity and harmonicity for a given interval. We will refer to that value as **total consonance** that is defined by the following formula:

$$\sigma(F, F') = \frac{\alpha(F, F') + \chi(F, F')}{2} \quad 5.6$$

Arithmetic mean is the simplest way to combine both contributions that will also give a value for total consonance between 0 and 1, where 1 indicates a perfect consonance between F and F' and 0 a complete lack of consonance. Now we can build what we will call affinitive tuning system $T_{\mathcal{A}}$ as a set of ordered pairs of affinitive intervals and their corresponding values of total consonance:

$$T_{\mathcal{A}}(F, F') = \{\{t, \sigma(F, tF')\} \mid t \in \mathcal{A}(F, F')\} \quad 5.7$$

As an example, Figure 5.1 depicts $T_{\mathcal{A}}(C4_6, C4_6)$ tuning. On horizontal axis we plot the value for interval t in cents and on vertical the value of total consonance $\sigma(C4_6, t \cdot C4_6)$. For convenience of determining to which note an interval corresponds to, in the bottom of the chart the piano keyboard is shown. Each key of the piano corresponds to the frequency ranges for notes from the Table 8.3. The red key highlights the note of musical context (in this case it is $C4$). One can see that each interval from Formula 5.5 is present in the graph and has a non-zero value of total consonance. The value of total consonance for the interval t corresponds to how consonant will be the total sound of F and tF' . For example, let's look at the interval of perfect fifth $3/2$ with $t = 702$ cents. The complementary set $F' = C4_6$ transposed on perfect fifth gives us the note $G4_6 = \frac{3}{2}C4_6$. Combining $tF' = G4_6$ and the context $F = C4_6$ will yield a value of total consonance $\sigma(C4_6, G4_6) = 0.44$. That means that if you add $G4_6$ to $C4_6$ the consonance will be 0.44. On the graph, the value for total consonance is shown as a sum of the contributions of harmonicity (orange lines) and affinity (green lines) stacked on top of each other. So, the value 0.44 for the perfect fifth is the total length of the vertical line (orange + green) and we see that it is just shy of 0.5.

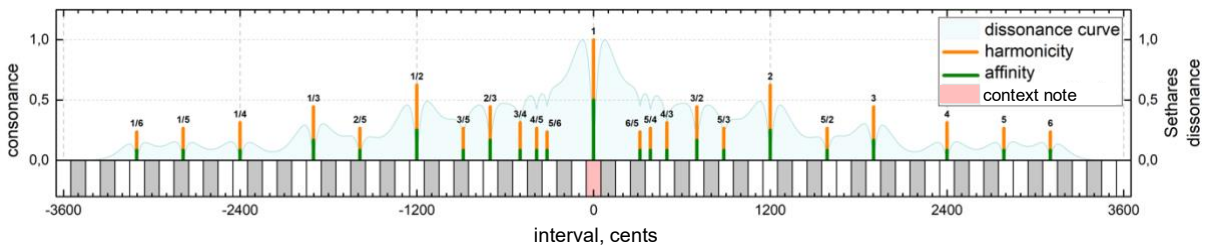


Figure 5.1 – Affinitive tuning $T_{\mathcal{A}}(C4_6, C4_6)$, where $C4_6 = 262 \cdot \mathbb{N}_6$. Black numbers represent exact numeric value of the interval t , orange and green bars show the values of harmonicity and affinity as parts of the total consonance σ . Blue curve is a Sethares dissonance curve for $C4_6$ spectrum. Note that sharp minima in dissonance curve mark the same intervals as in the affinitive tuning.

Below the graph the equally tempered piano keyboard is presented as a reference, with red note marking the note of a contextual set F .

Each interval of affinitive tuning coincides with a sharp minima of dissonance curve (blue curve) that is calculated for $C4_6$ spectrum. That is expected, as those minima of dissonance curve appear when frequencies of two sounds coincide, which is exactly the intervals at which affinity is greater than zero. The consonance values let us see which interval is most consonant relative to the musical context. The only perfectly consonant interval with $\sigma = 1$ is unison 1:1, followed by octaves 2:1 and 1:2 with $\sigma = 0.625$ and perfect fifths 3:2 and 2:3 with $\sigma = 0.44$, with other intervals having lower values.

The peculiar thing is that the tuning is symmetrical around the unison. That symmetry does not mean that the notes to the left of unison are the same as ones to the right, as one might assume. In fact, the interval $4/5$ marks $G\#3$, which is missing in the right half of tuning. It is important to note that affinitive tuning does not suppose nor predicts the octave equivalence³². That means that even though $G\#3$ is a part of the tuning, $G\#4$ is not. If we include octave equivalence as an additional axiom, we can transpose all the intervals into a single octave (Figure 5.2). Notice that after transposition the interval $4/5$ becomes $8/5$ which has a zero value of affinity.

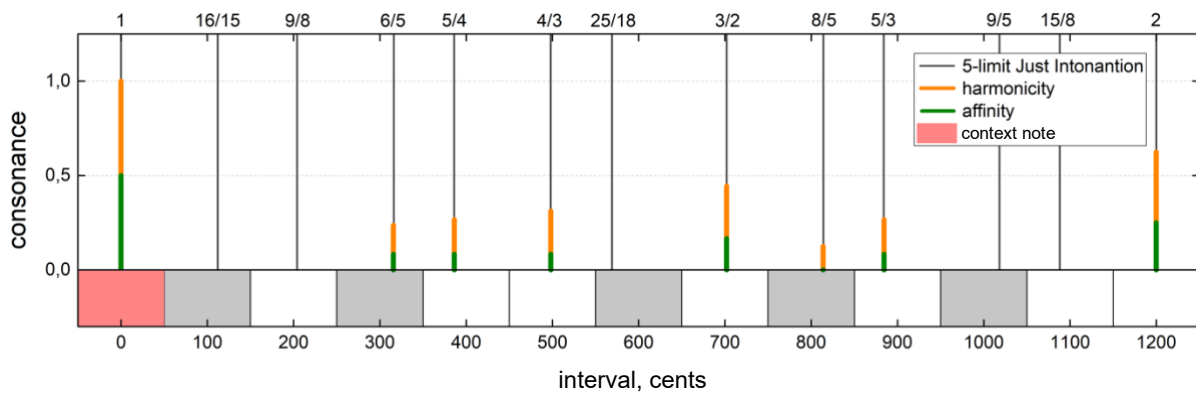


Figure 5.2 – The same affinitive tuning $T_{\mathcal{A}}(C4_6, C4_6)$ as in previous figure, but where all of the intervals are transposed to be within a range of a single octave. Black vertical lines mark the intervals of 5-limit³³ Just Intonation for 12 notes of chromatic scale. 5-limit intervals are chosen to be the simplest ratios that best approximate corresponding 7-limit intervals.

Comparing this tuning with 5-limit JI we see that JI includes affinitive tuning as a subset. The major differences between JI and affinitive tuning are the way a practitioner arrives to a particular set of intervals. In JI there are many ways one can come up with a set of intervals. For example, instead of using $25/18$ for augmented fourth, like in Figure 5.2, it is common to use the ratio

³² Octave equivalence refers to the observation that notes that are octave apart can be considered to be the same note.

³³ The limit of JI refers to the maximum prime number that can be used to construct ratios of intervals. In 5-limit JI the ratios of intervals are products of prime numbers up to 5, in 7-limit JI up to 7. For more details refer to [8].

45/32. In JI it is up to a practitioner to pick a set of intervals from all rational numbers. In affinitive tuning, the practitioner chooses sets F and F' , but the tuning is the result of a strict mathematical procedure.

Affinitive tuning is the product of two sets, F and F' . Any change in one of them will be reflected in the tuning. Let's investigate how affinitive tuning changes if the contextual set is an interval or a chord rather than a single note. We see in the Figure 5.3 that with every additional note in the contextual set F , the number of intervals in the tuning increases and the consonance values for each interval are changing. That shows the dynamic nature of tuning. As the musical piece progresses, the contextual set is changing, reflecting the changes in harmony and the affinitive tuning is changing with it. The maximum value of consonance reflects the consonance value of the current musical context, the more complex is the harmony the less is the value of consonance. The difference of the dynamism of affinitive tuning to the other dynamic tuning systems is that frequencies in contextual set F do not change as music progresses. Many other dynamic tuning

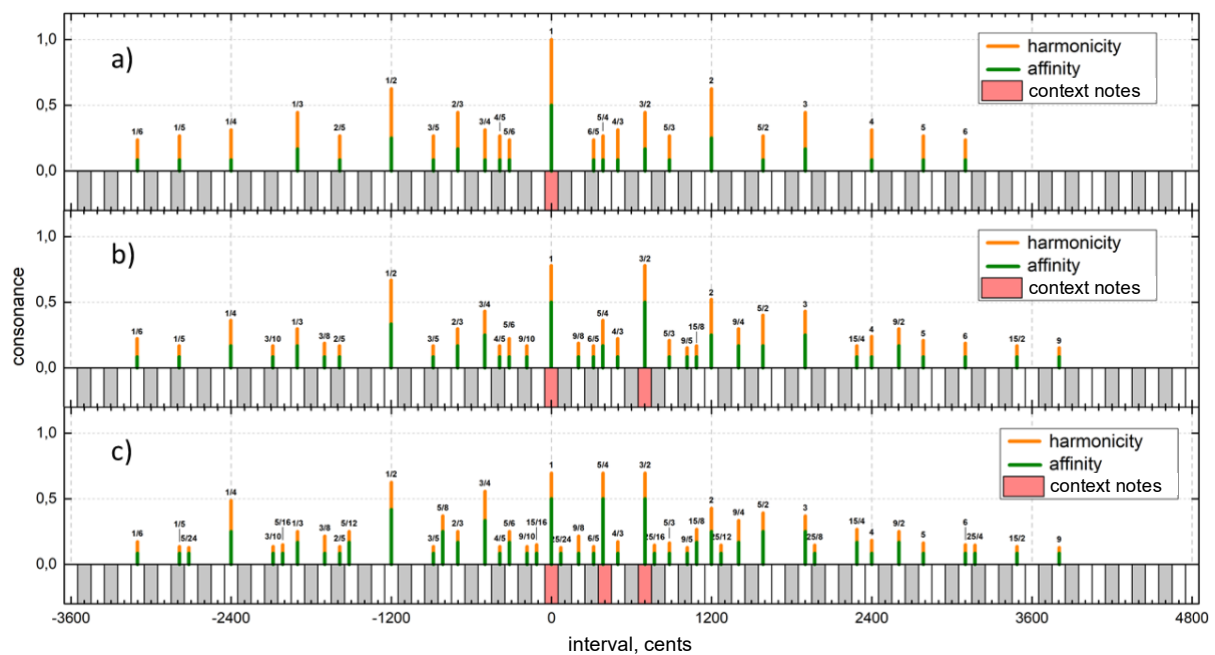


Figure 5.3 – Dynamic nature of affinitive tuning demonstrated with the change of contextual set F . On the a) set $F = C4_6 = 262 \cdot \mathbb{N}_6$. In b) $F = C4_6 \cup G4_6$ where $G4_6 = \frac{3}{2}C4_6$. In c) $F = C4_6 \cup E4_6 \cup G4_6$ where $E4_6 = \frac{5}{4}C4_6$.

systems, as described in [4], attempt to fix the 12-tet tuning, while keeping the interface of 12-tet keyboard.³⁴ Such tunings possess a class of artifacts when notes in the melody affect the pitch of

³⁴ The examples of such dynamic tunings would be a Sethares dissonance curve based tuning or a Hermodé tuning. Good review is found in [4]. Also, an interesting continuation of the Sethares work is Dynamic Tonality [13].

the notes in harmony, causing them to move up and down in order to minimize dissonance. Such situation does not occur in the affinitive tuning system, all frequencies in the context are fixed.

Similar dynamic behavior emerges in Sethares theory and free JI. The benefit of the affinitive tuning is very low computational requirements and the ability to quantify consonance of intervals using both affinity and harmonicity measures of consonance. Accounting for harmonicity does privilege harmonic sounds over inharmonic, which is evident from the Figure 5.4 that shows the affinitive tuning for highly inharmonic sound $F_{inharmonic}$

$$F_{inharmonic} = 262 \cdot \{1, 2.76, 5.41, 8.94, 13.35, 18.65\} \quad 5.8$$

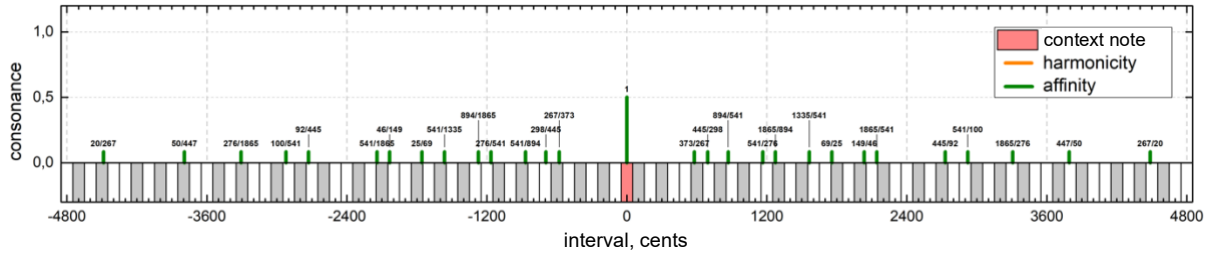


Figure 5.4 – Affinitive tuning for the inharmonic sounds $T_{\mathcal{A}}(F_{inharmonic}, F_{inharmonic})$. Very high degree of inharmonicity of $F_{inharmonic}$ reduces harmonicity contribution to near-zero values, which makes affinity a dominant contribution to consonance.

One can see that harmonicity contribution is practically non-existent, as it is orders of magnitude lower than affinity. In Sethares framework there is no differentiation or ranking between different timbres. All of them are equally usable and depend on the taste of the practitioner. In suggested framework there is a visible differentiation between timbres and tunings. Harmonic sounds as in Figure 5.1 seem to produce much richer and expressive tuning than inharmonic sounds in Figure 5.4. That is evident from a greater variance of consonance values between different intervals. Other inharmonic timbres can be more expressive than $F_{inharmonic}$, but with set theoretic approach we have an objective measure on how we can judge it.

In the case of affinitive tuning, harmonicity plays an important role, however, it does not affect which intervals are included in tuning. It is possible that some meaningful intervals with large consonance values may be missed out if we only use affinity to determine intervals. The most glaring issue that indicates that we have to find a way to include harmonicity into interval derivation is that if sets F and F' have only 1 frequency, the tuning will consist of only a single interval. As we will see from the next chapter, that misses a lot of intervals with large values of consonance. Thus, let's investigate which tuning emerges from the harmonicity metric.

5.2 Harmonic tuning system

Just like affinity, harmonicity is a function of contextual set F and complementary set F' . To get the affinitive tuning, we came up with a procedure of finding all intervals t for which the value of affinity $\alpha(F, tF')$ is greater than zero (Formula 5.1). Such approach generated a tuning with a moderate number of intervals, which is important as having too little intervals is not very expressive, and too much is overwhelming. Unfortunately, the same approach does not work as well for harmonicity. The reason for that is that harmonicity is strictly zero when a set of frequencies is incommensurable (see Chapter 4.3). That means that if one of the sets F and F' is incommensurable, the harmonicity will be 0 for all intervals t . If on the other hand both sets F and F' are commensurable, then harmonicity will be 0 for all irrational t and greater than zero for all rational t .

The same conclusion, that every rational number is to some degree consonant, was reached by theoreticians of JI. The degree of consonance in JI is measured as a simplicity of the ratio that constitutes that interval [8]. Let's investigate the distribution of harmonicity in the simplest case when both F and F' have a single element and equal to $C4_1 = 262 \cdot \mathbb{N}_1$. As we cannot check all rational numbers, we have to come up with some arbitrary limitations. Let's calculate a total consonance for all rational numbers with denominator less than 60 and in the span of ± 3 octaves (Figure 5.5).

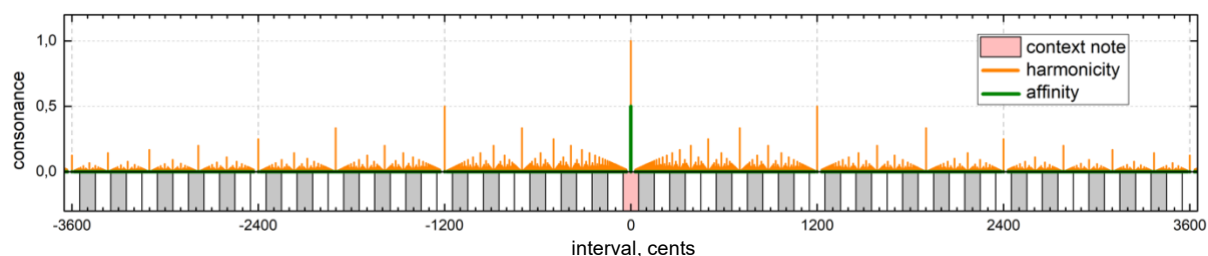


Figure 5.5 - Distribution of harmonicity for a span of intervals ranging from -3 to +3 octaves. The intervals are chosen so that the largest value of denominator is 60. Consonance is calculated for $F = F' = 262 \cdot \mathbb{N}_1$ being a single partial with a frequency that corresponds to the note $C4$. Due to the high density of intervals, we omit the labels for their ratios.

As seen from the graph, each value of harmonicity is a local maximum surrounded by much smaller values. That means that tiny deviations from precise ratios cause dramatic reduction of harmonicity.³⁵

The graph in Figure 5.5 demonstrates an issue with affinitive tuning that was stated at the end of the previous chapter. For sets with a single partial like $C4_1$, only the unison 1:1 has a non-zero value of the affinity, while there are plenty of intervals with large values of harmonicity to pick from. What is remarkable is the beautiful fractal structure of consonance that can be described by a modified Thomae's function³⁶:

$$\tau_m(t) = \begin{cases} \frac{1}{\max(p, q)} & \text{if } t = \frac{p}{q} \text{ (} t \text{ is rational), with } p \in \mathbb{Z} \text{ and } q \in \mathbb{N} \text{ coprime} \\ 0 & \text{if } t \text{ is irrational} \end{cases} \quad 5.9$$

The Figure 5.6 demonstrates that graph of Formula 5.9 (blue lines) coincides with the distribution of total consonance in Figure 5.5 (orange lines in Figure 5.6), meaning that it is the same function.

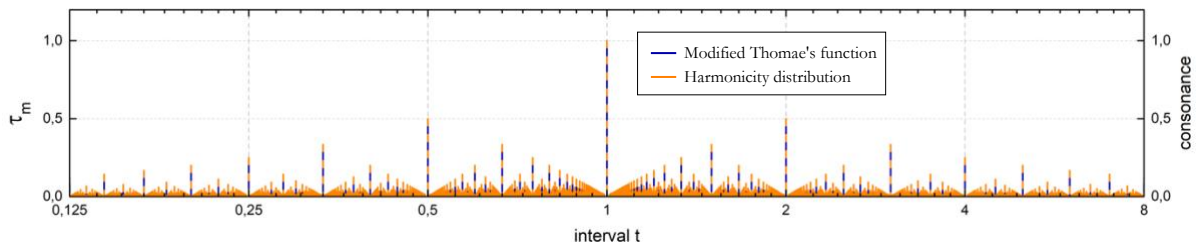


Figure 5.6 – Comparison of the distribution of total consonance for $F = F' = C4_1 = 262 \cdot \mathbb{N}_1$ (orange lines) on the right axis with modified Thomae's function described by equation 5.9 (blue line) on the left axis.

This non-trivial result can be proven algebraically using the definitions of harmonicity and affinity which is done in Chapter 8.5.

The Formula 5.9 proves the definition of consonance used by JI theoreticians that consonance of an interval is inversely proportional to the complexity of its ratio. Complexity of the ratio in the Formula 5.9 is represented by the value $\max(p, q)$. Thus, the interval is more complex the larger

³⁵ Floating point arithmetic proved to be unsuitable for plotting a harmonicity distribution. Tiniest deviations from the exact ratios, such as $4/3$, caused a near zero value of harmonicity. Thus, a rational number arithmetic using Fraction.js library [14] had to be used to derive values of intervals.

³⁶ See Chapter 8.4 for more info on Thomae's function.

the number is needed to represent it, this way perfect fifth 3:2 is more consonant than major third 5:4 and the direction of interval does not influence consonance rating³⁷. Also notice that this result is true for the total consonance that includes affinity. Distribution of harmonicity does not follow Formula 5.9, as it has the same values of octave and unison as seen in the Figure 5.7.

This result is true only for the sounds with a single partial. More complex sounds change the harmonicity distribution. The Figure 5.7 compares the harmonicity distribution for harmonic sounds with 1, 6, and 256 partials. One can see that there is a noticeable difference in harmonicity values between cases of 1 and 6 partials, but increasing the number of partials beyond 6 does not change the distribution too much.

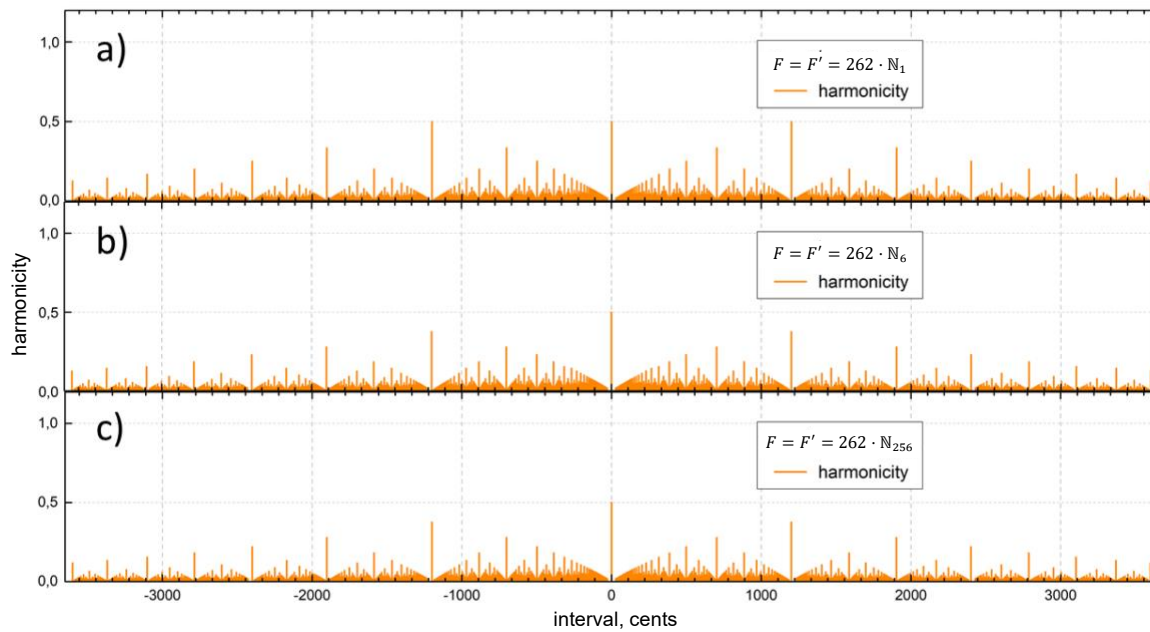


Figure 5.7 – The change in harmonicity distribution with increasing number of harmonics, affinity values are omitted. a) Harmonicity distribution for a single partial $F = F' = 262 \cdot N_1$. b) Harmonicity distribution for harmonic sound with 6 partials $F = F' = 262 \cdot N_6$. c) Harmonicity distribution for harmonic sound with 256 partials $F = F' = 262 \cdot N_{256}$. Increasing number of partials past 6 does not lead to dramatic change in harmonicity distribution.

More noticeable changes to harmonicity distribution happen when more notes are added to the context set. The Figure 5.8 demonstrates what will happen with the distribution of harmonicity if we start adding additional notes to the context, starting with a single $C4_6 = 262 \cdot N_6$.

³⁷ For example, the perfect fifth built up from C4 to G4 is represented by the interval 3:2, while if built down from C4 to F3 is represented by 2:3. Both are perfect fifth and both will have the same complexity $\max(3,2) = \max(2,3)$.

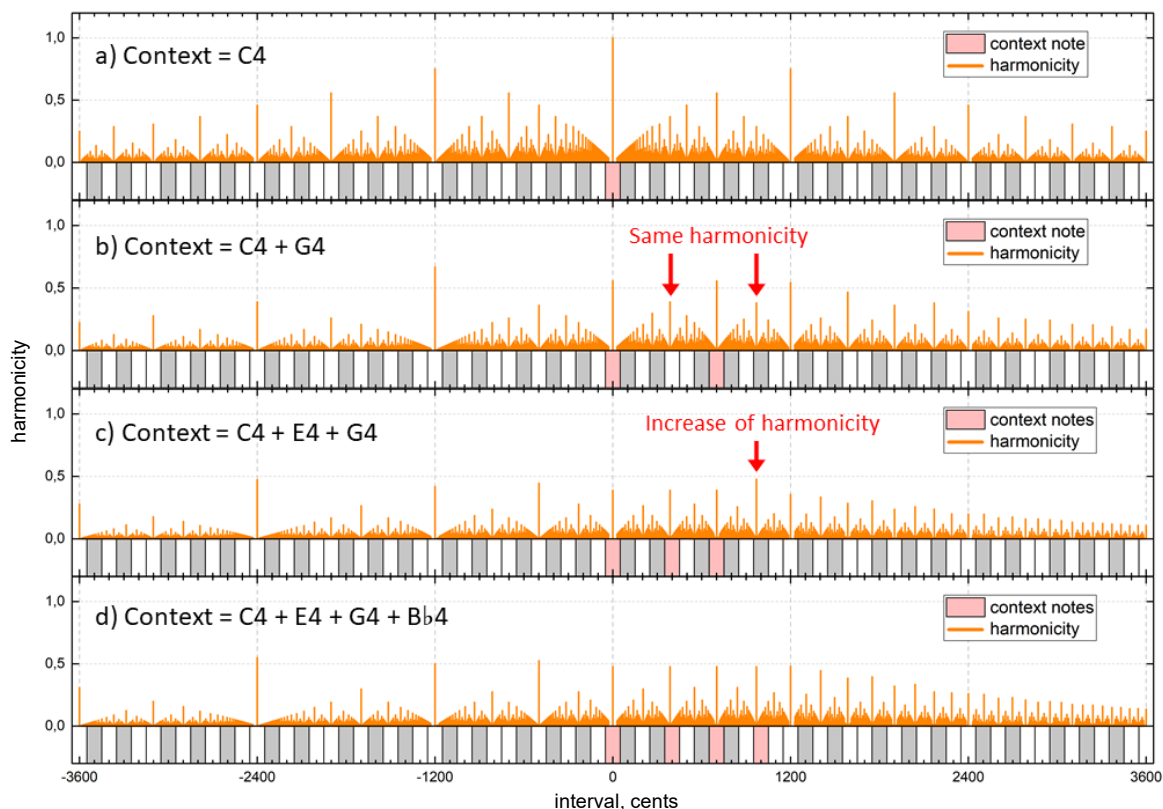


Figure 5.8 – Changes in harmonicity distribution caused by changes in contextual set, affinity values are omitted. a) The context is a single note $F = C4_6$. b) The context is an interval of perfect 5th $F = C4_6 \cup G4_6$. Note the same values for harmonicity for major 3rd 5:4 and harmonic 7th 7:4. c) The harmonicity distribution for major triad $F = C4_6 \cup E4_6 \cup G4_6$. Note the increase in harmonicity for $Bb4_6$. d) The harmonicity distribution for $C7$ chord $F = C4_6 \cup E4_6 \cup G4_6 \cup Bb4_6$.

When considering what note to add to a single $C4_6$, the most consonant interval (as measured by harmonicity alone) is a perfect fifth up which gives $G4_6 = \frac{3}{2} \cdot C4_6$. Adding it to the context leads to the surprising result of notes $E4_6 = \frac{5}{4} \cdot C4_6$ and $Bb4_6 = \frac{7}{4} \cdot C4_6$ having the same value of harmonicity. Adding $E4_6$ to context to form a major triad further emphasizes the harmonic seventh $Bb4_6$ as a note that increases the overall harmonicity of a chord! That runs contrary to the usual notion of major triad being more consonant than major 7th chord. One might take it as a proof that a reliance on harmonicity as a sole measure of consonance is error prone. As shown in Figure 5.9, adding affinity contribution changes consonance distribution in favor of common triadic harmony.

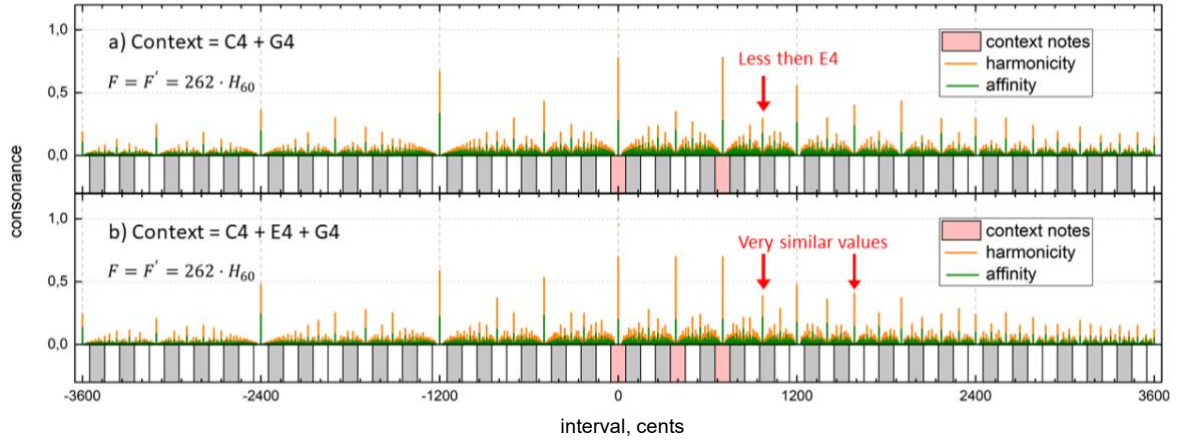


Figure 5.9 – Distribution of total consonance for a rich harmonic sound with 60 partials to demonstrate a preference for triadic harmony that happens when including affinity contribution. The increase in partial count is done to equalize contributions from harmonicity and affinity. Lower number of partials does not change the result. a) Harmonicity distribution for two notes in the context $F = C4_6 \cup G4_6$, comparing to Figure 5.8 the harmonic 7th does have a smaller value than major 3rd. b) Harmonicity distribution for the major triad in the context. Harmonic 7th does not increase consonance as in Figure 5.8. However, the consonance value of $E5_6$ is the same as for harmonic 7th $Bb4_6$.

As shown in previous chapter, affinity becomes dominant contribution to consonance if dealing with inharmonic sounds. The Figure 5.10 shows the harmonicity distribution for inharmonic sets like in the Formula 5.8. One can see that the overall shape is very similar to the case of a single partial in Figure 5.7, but with the values of harmonicity several orders of magnitude smaller.

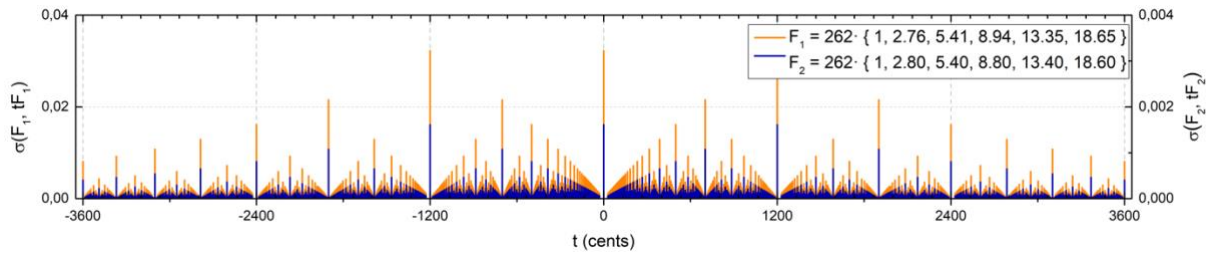


Figure 5.10 – Comparison of harmonicity distribution of 2 inharmonic sounds. On the right axis - harmonicity for inharmonic sound $F_{inharmonic}$ described by Formula 5.8. On the left axis – harmonicity for the inharmonic sound made by rounding off the last digit of partials in $F_{inharmonic}$. Notice that rounding off a single digit increases harmonicity by almost 20 times.

As mentioned previously, harmonicity is very sensitive to the precision of frequencies in sets. Rounding off a single digit can lead to dramatic increase of harmonicity of the sound, as shown in Figure 5.10.

Because harmonicity naturally prefers rational numbers as described by small ratios, one can say that JI is the natural harmonic tuning we were seeking to find in this chapter. That is true if we add octave equivalence as an additional axiom. Also, having a measure of consonance provides more clarity than JI to the consonance of various note combinations and gives us a new way we can limit the number of intervals in the tuning. For example, we can only include the most

consonant intervals that have a total consonance higher than some number h . This way, the set of harmonic intervals will be determined by the following formula:

$$\mathcal{H}(F, F', h) = \{t \mid \chi(F \cup tF') > h\} \quad 5.10$$

That lets us define the harmonic tuning in the following way:

$$T_{\mathcal{H}}(F, F', h) = \{\langle t, \sigma(F, tF') \rangle \mid t \in \mathcal{H}(F, F', h)\} \quad 5.11$$

The Figure 5.11 shows that harmonic tuning suits better for a sound with few partials than the affinitive tuning, given a carefully picked value of h . Having too small of value for h generates too much intervals in Figure 5.11.b, but raising it to $h = 0.23$ filters out most of intervals leaving only the most consonant ones (Figure 5.11.c).

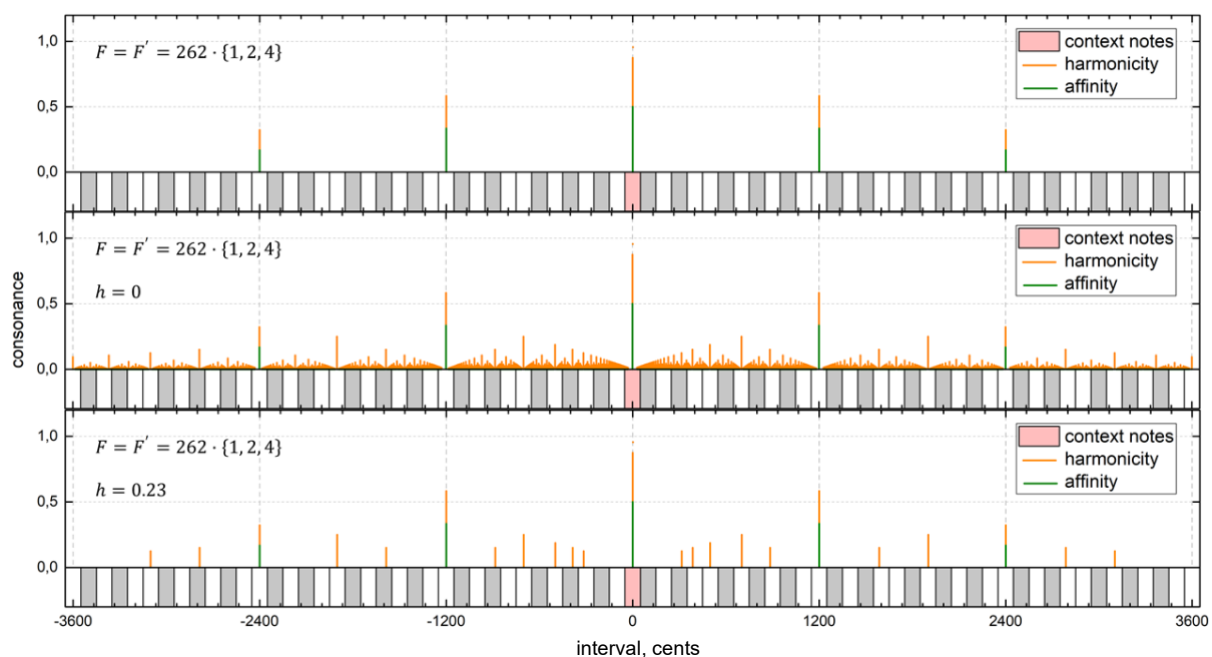


Figure 5.11 – Advantage of harmonic tuning for sounds with small number of partials. a) Affinitive tuning for $F = F' = 262 \cdot \{1, 2, 4\}$. Tuning is useless for music making, as it has only the octave and double octave intervals. b) Harmonic tuning with $h = 0$ for the same set as in a). It has too many intervals due to the low value of h . c) Harmonic tuning with $h = 0.23$ for the same sound as in a) and b). This value of h filters out most of the intervals leaving us with perfectly usable tuning.

Defined in this way, harmonic tuning suits more to the sounds with few partials than the affinitive tuning. That advantage, however, comes with the much larger computational overhead, as we need to process hundreds or thousands of intervals for their consonance values to be able to pick the

most consonant ones. Another drawback is a difficulty to use harmonic tuning for inharmonic sounds, as some intervals with non-zero affinity can be missed due to insufficient precision (Figure 5.14.a). The last issue is a very high sensitivity to deviation from simple ratios which contradicts empirical observations. I believe it is possible account for human auditory system in various ways to alleviate that problem, however, doing that is not in scope of this article.

Just like affinitive tuning, harmonic tuning turned out to have its own set of advantages and disadvantages. The question that arises, is there another way to include harmonicity in tuning generation?

5.3 Harmonic superset tuning system

In Chapter 4.3 we discussed how different frequencies of harmonic set form a single pitch. We also discussed that any commensurable set F has a harmonic superset $H(F)$, which means that F is a part of a single pitch $H(F)$. That harmonic superset is a virtual pitch of that sound. In that sense, we may say that two notes comprising the major 3rd interval in Figure 4.4 are two separate pitches, but at the same time we can say that together they form a part of a single 88Hz pitch. This way instead of finding affinitive intervals for the sets F and F' , we can find them for the corresponding harmonic supersets $H(F)$ and $H(F')$.

Using supersets to generate intervals should provide an improvement for harmonic sounds with missing partials like in Figure 5.11.a. Superset will fill the gaps in the spectrum and provide more intervals to tuning. However, if the sound consists of only a single frequency, harmonic superset, as defined by the Formula 4.2, will also have only 1 frequency. That can be fixed, as the condition $k = \frac{\max(F)}{\gcd(F)}$ that limits number of frequencies in harmonic superset, is the minimal condition that guarantees that all frequencies in F are included in $H(F)$. Thus, if k is too small to get a meaningful tuning, we can increase it on a number n to get more intervals:

$$H_n(F) = \gcd(F) \mathbb{N}_{k+n}, \quad k = \frac{\max(F)}{\gcd(F)} \quad 5.12$$

To define the harmonic superset tuning system, we need to assign the values of consonance to each interval. We can measure consonance for either harmonic supersets $H_n(F)$ and $H_m(F')$ or the original sets F and F' . I would argue that since frequencies in supersets are not present in the

physical sound, they should not be treated equally to the real frequencies. Thus, the total consonance should be calculated for F and F' and not their supersets. Given that, we can define harmonic superset tuning system with the following formula:

$$T_S(F, F', n, m) = \left\{ \langle t, \sigma(F, tF') \rangle \mid t \in \mathcal{A}(H_n(F), H_m(F')) \right\} \quad 5.13$$

Arguments n and m can be used to increase the number of partials in harmonic supersets to provide a tuning for sets with a single partial, as shown in Figure 5.12.

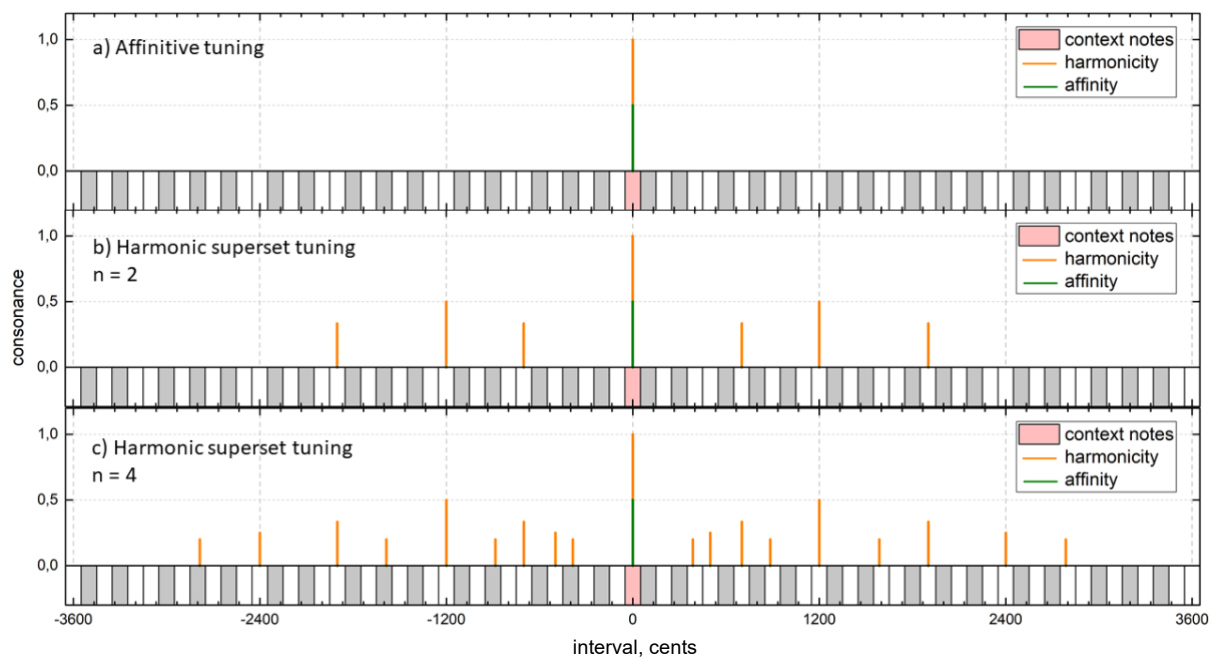


Figure 5.12 – Demonstration of the advantage of harmonic superset tuning system over the affinitive tuning system for the sounds with a single partial $F = F' = 262 \cdot \mathbb{N}_1$. a) Affinitive tuning for the sounds with a single frequency. b) Harmonic superset tuning with $n = m = 2$ and a few intervals. c) Harmonic superset tuning with $n = m = 4$ and a usable tuning that contains all of the most consonant intervals. Further increase of parameters n and m will lead to even richer sets of intervals.

Notice that most of the intervals (apart from unison 1:1) in Figure 5.12 have zero value for affinity. That is expected, as consonance is calculated for sets F and F' , while intervals are generated from their supersets $H_n(F)$ and $H_m(F)$. Using more complex sounds will yield more intervals with non-zero affinity.

Affinitive tuning is a subset of harmonic superset tuning, because the condition for k in the Formula 5.12 guarantees that all partials in F and F' are included in supersets and thus, all the

intervals they generate will be included in harmonic superset tuning. That is seen in Figure 5.13, where change in contextual set F causes similar changes in tuning as affinitive tuning in Figure 5.3. However, in contrast to affinitive tuning, we see additional intervals with 0 value of affinity in harmonic superset tuning.

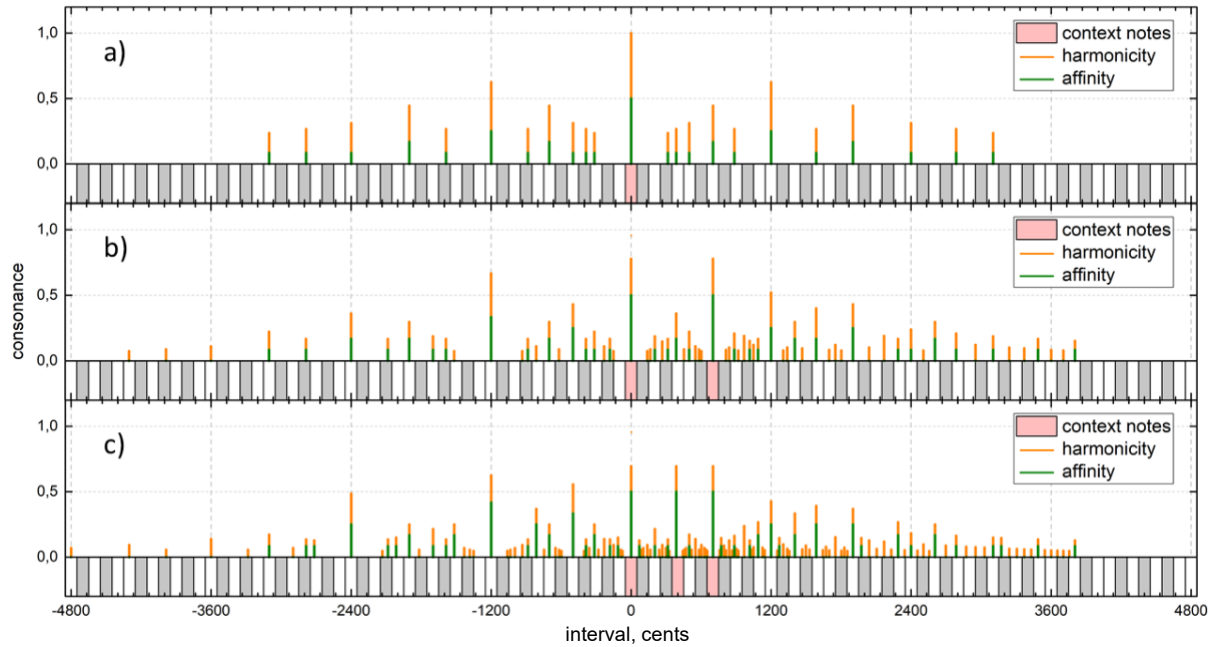


Figure 5.13 – Demonstration of the dynamic nature of harmonic superset tuning from the change of the context set. a) Harmonic superset tuning for the contextual set being a single note $F = C4_6 = 262 \cdot \mathbb{N}_6$. The tuning is identical to affinitive tuning in the Figure 5.3.a b) Harmonic superset tuning for the interval of fifth $F = C4_6 \cup G4_6$. We notice additional intervals with 0 affinity that are missing in affinitive tuning in Figure 5.3.b c) Harmonic superset tuning for the major triad $F = C4_6 \cup E4_6 \cup G4_6$. Comparison with the Figure 5.3 reveals that affinitive tuning is a subset of harmonic superset tuning.

With the addition of 5^{th} to the contextual set we get an extra low octave compared to affinitive tuning. That happens because with each addition of a new note, the virtual pitch of contextual set F drifts lower and lower, the number of partials in $H_n(F)$ increases more and more to satisfy minimal partial requirement. That leads to more intervals with zero value for affinity but non-zero harmony value, resulting in a richer tuning.

Because harmony contribution is so sensitive to small deviations of partials (Figure 5.10), the affinitive tuning may be a better fit to inharmonic sounds as it does not spend extra computation resources on the intervals with dismissively small consonance values. However, compared to the harmonic tuning there is no risk to miss valuable intervals as harmonic superset tuning includes all intervals from affinitive tuning. The graphs in Figure 5.14 clearly demonstrates that advantage. Harmonic superset tuning does not depend on precision parameter to capture all of the important intervals from both affinitive and harmonic perspectives.

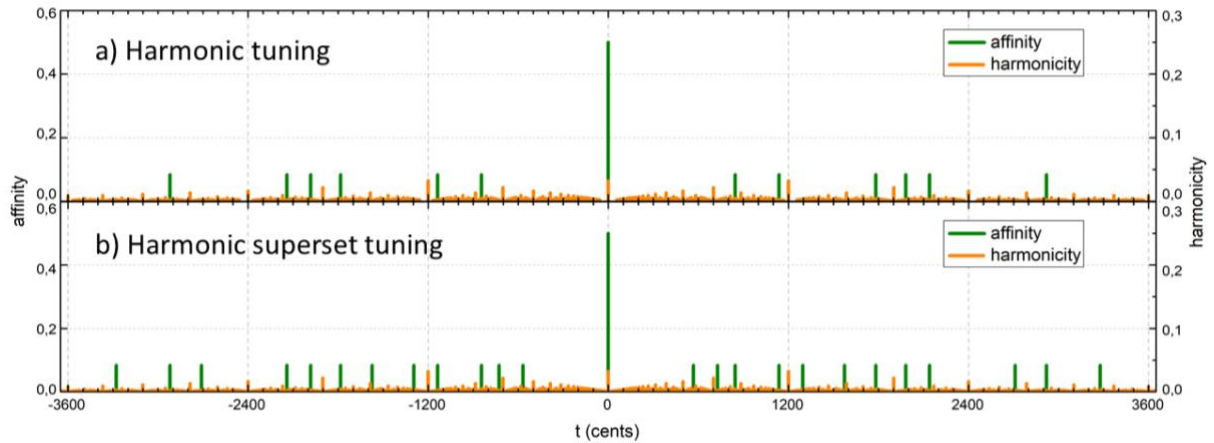


Figure 5.14 – Demonstration that harmonic superset tuning does not miss important intervals for inharmonic sounds as does harmonic tuning. a) Harmonic tuning with a set of intervals limited by maximum value of denominator of 30 for the sets $F = F' = 262 \cdot \{1, 2.8, 5.4, 8.8, 13.4, 18.6\}$. b) Harmonic superset tuning for the same sets as a). Notice additional intervals with non-zero affinity that are missing in harmonic tuning in a). On both graphs the values for harmonicity are scaled up two times to be visible and are plotted against the right axes.

Harmonic superset tuning improves upon previous two systems by employing affinity and harmonicity in both aspects of tuning system: generation of intervals and measurement of consonance. Due to that, it improves upon the drawbacks of the other systems, while not losing the benefits. It is a safe tuning system to pick that will not miss on any important intervals and that allows the greatest flexibility for composition. Affinitive tuning is a good alternative that should be considered to reduce computation needed to calculate the tuning.

5.4 Discussion

In this chapter I suggested three different tuning systems that naturally emerge from affinity and harmonicity consonance measures. Affinity generates affinitive tuning that is similar to the system based on dissonance curves suggested by Sethares in [4]. Harmonic tuning that emerges from harmonicity gives preference for simple ratio intervals resembling JI [8]. Both set theoretic systems improve upon mentioned alternatives, as for each interval they assign a measure of total consonance. That is an important contribution, as that measure accounts for two of the most prevalent contributions to the perception of consonance: spectral interference and harmonicity/periodicity.³⁸

³⁸ According to [1], spectral interference and harmonicity/periodicity plays the largest role in the perception of consonance. Affinity, being a proxy measure for spectral interference, and harmonicity, a measure of harmonicity/periodicity, are both included in the total consonance that is defined by the Formula 5.6.

However, that improvement is not enough to check all the boxes in the Table 2.1. The affinitive tuning may improve upon Sethares tuning by reducing computation and accounting for harmonicity of spectrum, but just like Sethares theory, it does not generate a usable tuning for sounds with few partials. The similar thing is true for harmonic tuning. Just like JI, it does not provide a good tuning for inharmonic timbres, even though consonance rating may identify the intervals with large value of affinity, it does not provide a guarantee that all such intervals will be found.

It is a non-trivial task to combine spectral interference and harmonicity/periodicity into a single framework to generate tuning. To the best of my knowledge, it has never been done to this day.³⁹ Luckily, the set theoretic approach provides almost trivial solution for that problem. By replacing original sets with corresponding harmonic supersets (corresponding virtual pitches), it is capable of generating tuning that is suitable for all types of sounds in the Table 2.1 – harmonic superset tuning. If true, that makes harmonic superset tuning the most broadly applicable tuning theory known to date.

On this note, it can be concluded that the aim of the paper is satisfied. The harmonic superset tuning system is a good candidate for a natural tuning system. It originates in pure mathematics and does not depend on subjective preferences. Potentially, its applicability is broader than any other tuning system which satisfies Ptolemy condition for validity of idealistic theory. However, more investigation has to be made to back up that claim. One way of doing that, is to test the system in practice and see what fruits come out of it.

The practical application necessarily raises the question of the appropriate interface to that tuning. We used a piano keyboard as a reference to show which interval corresponds to each note, but it does not suggest that using piano keyboard is in any sense appropriate. It is clear from Figure 5.14 that harmonic superset tuning is microtonal, and the density of intervals builds up quite quickly. It is possible to reduce the complexity to be compatible with the keyboard layout, but it may not be a good way forward. Like it was discussed in Chapter 2.3, the piano keyboard is a matched interface for 12-tet. Thus, moving away from 12-tet implies leaving the keyboard layout. I see the description and creation of such interface as my next endeavor, and I believe the digital technologies should make it accomplishable.

³⁹ In the work [1] authors claim to combine spectral interference and harmonicity into a single framework, however, the purpose of that framework was to explain the consonance preferences of people and to provide a tuning system for musicians to use.

6 Conclusion

On that note I would like to end the discussion of tuning systems and call the reader for further investigation and experimentation with set theoretic approach to musical tuning. We went through a history and philosophy that surrounds the tuning theory and postulated the definition of consonance. We demonstrated how that definition can be captured by mathematics of sets by defining two terms. The first one is affinity; it is responsible for measuring how well different notes fit together. The second one is harmonicity; it measures how multiplicity of different pitches forms a unity of one larger pitch. Those two sides of consonance naturally give birth to affinitive and harmonic tuning systems, each with its own set of advantages and disadvantages. We also saw how marrying affinity and harmonicity together produces harmonic superset tuning system that has the broadest limits of applicability of any other theory and serves as a candidate for a natural tuning system. All the tuning systems we discovered are products of pure mathematics. Thus, their existence does not depend on human auditory system or empirical observations.

I do not consider this work as the final and comprehensive analysis of how set theory can be used to derive tuning systems. This is just my take on what can be done, a steppingstone to build upon. Can affinity and harmonicity be defined on a deeper level that will strengthen the theoretical basis? I am sure they can. Can there be other aspects to consonance, sound or tuning that I never thought of? Most certainly there are. Is it possible to build better tuning systems using the same definitions for consonance contributions? No doubt, it is. While writing this paper I stumbled upon things I never expected to encounter. The unexpected links to other tuning theories, the simplicity of math and beauty of graphs - all prove to me that the set theoretic approach does reflect the nature of musical tuning in a deep sense. That inspires me to continue the work on that topic by bringing that theory to practice. I don't know where that will lead me, but hopefully to some good music.

Follow my work on New Tonality YouTube channel [9] and website [10]. Big thank you to all supporters on Ko-fi and all viewers of the channel. I express my gratitude to the authors of the cited papers and everybody else who shared their love and knowledge on tuning theory and music. I am thankful to all my teachers who brought me up and taught me everything I know. I am especially thankful to my wife for the enormous help of editing and proofreading this paper. I dedicate this work to my family: my wife Ekaterina, my mother Liubov and my brother Yaroslav. Most of all I praise the Holy Trinity for the gifts of music, beauty and reason.

Glory to the Lord!

7 Bibliography

- [1] P. M. C. Harrison and M. T. Pearce, "Simultaneous Consonance in Music Perception and Composition," *Psychological Review*, pp. 216-244, 2020.
- [2] H. L. F. Helmholtz, *On the Sensations of Tone as a Physiological Basis for the Theory of Music*, Cambridge University Press, 1875.
- [3] A. Burov and L. Burov, "Genesis of a Pythagorean Universe," in *Trick or Truth? The Frontiers Collection*, Springer, Cham, 2016, pp. 157 - 170.
- [4] W. A. Sethares, *Tuning, Timbre, Spectrum, Scale*, Springer London, 2005.
- [5] R. Plomp and W. J. M. Levelt, "Tonal Consonance and Critical Bandwidth," *J. Acoust. Soc. Am.* 38, pp. 548 - 560, 1965.
- [6] K. Z. Gill and D. Purves, "A Biological Rationale for Musical Scales," *PLOS ONE*, 2009.
- [7] D. Bowling, D. Purves and K. Gill, "Vocal similarity predicts the relative attraction of musical chords," *Psychological and Cognitive Sciences*, vol. 115, no. 1, pp. 216-221, 2017.
- [8] K. Gann, *The Arithmetic of Listening Tuning Theory and History for the Impractical Musician*, University of Illinois Press, 2019.
- [9] V. V. Deriushkin, "New Tonality YouTube channel," [Online]. Available: https://www.youtube.com/@new_tonality.

- [10] V. V. Deriushkin, "New Tonality Website," 2025. [Online]. Available: <https://newtonality.net/>.
- [11] R. E. Goldberg, *Where Nature and Art Adjoin: Investigations into the Zarlino-Galilei Dispute, Including an Annotated Translation of Vincenzo Galilei's Discorso intorno all'opere di Messer Gioseffo Zarlino*, Indiana University, 2011.
- [12] H. Partch, *Genesis Of A Music: An Account Of A Creative Work, Its Roots, And Its Fulfillments*, Hachette Books, 1979.
- [13] A. J. M. W. S. J. Plamondon, "Dynamic Tonality: Extending the framework of tonality into the 21st century," *Proceedings of the Annual Conference of the South Central Chapter of the College Music Society*, 2009.
- [14] R. Eisele, "Fraction.js," [Online]. Available: <https://github.com/rawify/Fraction.js>.
- [15] Ptolemy and J. Solomon, *Ptolemy Harmonics. Translation and Commentary*, Brill, 2000.
- [16] Aristoxenus and H. S. Macran, *The harmonics of Aristonexus*, Oxford : At the Clarendon Press, 1902.
- [17] Britannica, "Pythagoreanism," [Online]. Available: <https://www.britannica.com/topic/number-symbolism/Pythagoreanism>. [Accessed 9 5 2025].

8 Supplementary materials

8.1 Common set theoretic operations and symbols used in this article

The following is the list of common set theoretic operations and symbols with description and examples. Those operations are not tuning theory specific and are the same as in the mathematical literature.

Table 8.1 - Common symbols and operations on sets used in this article.

Operation / Symbol	Notation	Description
Size (cardinality)	$ A $	Number of elements in the set
Empty set	$\emptyset$	A set that has no elements and has a zero size $ \emptyset = 0$.
Element of set, Not an element of set	$x \in A$, $x \notin A$	For example, $x \in A$ is read as “ x is an element of the set A ”; and $x \notin A$ is read as “ x is not an element of the set A ”.
Subset, Proper Subset	$A \subseteq B$, $A \subset B$	For two sets A and B , A is a subset of B if every element in A is also in B . For example, if $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ then $A \subseteq B$. That definition is also true if $A = B$. Sometimes it is necessary to say that A is a subset of B but not equal to B . In that case A is called a proper subset of B and is written as $A \subset B$.
Superset, Proper Superset	$A \supseteq B$, $A \supset B$	If A is a subset of B , then B is a superset of A . For example, $B = \{1, 2, 3\}$ is a superset of $A = \{1, 2\}$. Similarly to the definition of proper subset, when $B \neq A$ we call B a proper

		superset of A . For example, a set of whole numbers is a proper superset of natural numbers $\mathbb{Z} \supset \mathbb{N}$.
Union	$A \cup B$	Union of two sets A and B is a set that contains all the elements that are in A or in B or in both A and B . Formal definition $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$. For example, if $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, then the union $A \cup B = \{1, 2, 3, 4\}$. Note that number 3 is not repeated as by definition, elements in a set are always unique.
Intersection	$A \cap B$	Intersection of two sets A and B is a set that contains all the elements that are shared between A and B . Formal definition looks following $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$. For example, if $A = \{1, 2, 3\}$ and $B = \{3, 4\}$ then intersection $A \cap B = \{3\}$.
Ordered pair	$\langle a, b \rangle$	The elements in a set are unordered, meaning that $\{a, b\}$ and $\{b, a\}$ is the same set. Sometimes it is helpful to keep the order of elements. Ordered pair $\langle a, b \rangle$ is not the same as $\langle b, a \rangle$.

8.2 Tuning theory specific set theoretic operations and symbols

Most of the labeling of set theoretic operations used in this article follows the usual convention. However, to perform common tasks within a tuning theory, such as transposition on an interval, required adding some new operations and labels that are not common in broader literature. The following table is the list of such operations that are unique to this article.

Table 8.2 - The list of tuning theory specific operations and symbols used in this article

Transposition on an interval (scalar multiplication)	tF or $t \cdot F$	Defines an operation of transposition of the set F on an interval t . The transposition is done by multiplying each frequency in F on t . Formal definition: $tF = \{tf \mid f \in F\}$. For example, to transpose a set $F = \{1, 2, 3\}$ up an octave $t = 2$, we can write $2 \cdot F = \{2, 4, 6\}$. This operation can also be called a scalar multiplication.
Elementary harmonic sets	$\mathbb{N}_k$	The symbol $\mathbb{N}$ refers to the set of all natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$, the index k is used to limit the number of elements in set the following way $\mathbb{N}_k = \{n \mid n \in \mathbb{N} \text{ } n \leq k\}$. For example, $\mathbb{N}_3 = \{1, 2, 3\}$. Such sets are handy to represent harmonic sounds by transposing them on the fundamental frequency (see Formula 3.4).
Least common multiple	$\text{lcm}(F)$	Least common multiple of a set F is the smallest number that is divisible without remainder on every element of the set F . For example, $\text{lcm}(\{10, 15\}) = 30$ as $\frac{30}{10} = 3$ and $\frac{30}{15} = 2$.

Greatest common divisor	$\gcd(F)$	Greatest common divisor of a set F is the largest number to which every element of the set F is divisible without remainder. For example, $\gcd(\{10, 15\}) = 5$ as $\frac{10}{5} = 2$ and $\frac{15}{5} = 3$.
Note notation	$A4_6$, $C\#5_3$	This notation is used to indicate the musical note the set refers to and the number of harmonics in that set. For example, the set $A4_6$ stands for a note $A4$ with 6 harmonics.

8.3 Frequencies of notes in standard 440Hz tuning

Table 8.3 – Frequency ranges of musical notes

C0	15.88 – 16.82 Hz	C2	63.54 – 67.32 Hz	C4	254.18 – 269.29 Hz
C#0 / Db0	16.82 – 17.82 Hz	C#2 / Db2	67.32 – 71.33 Hz	C#4 / Db4	269.28 – 285.3 Hz
D0	17.82 – 18.88 Hz	D2	71.32 – 75.57 Hz	D4	285.3 – 302.26 Hz
D#0 / Eb0	18.89 – 20.01 Hz	D#2 / Eb2	75.56 – 80.05 Hz	D#4 / Eb4	302.27 – 320.24 Hz
E0	20.01 – 21.2 Hz	E2	80.06 – 84.82 Hz	E4	320.24 – 339.28 Hz
F0	21.2 – 22.46 Hz	F2	84.82 – 89.86 Hz	F4	339.28 – 359.46 Hz
F#0 / Gb0	22.46 – 23.79 Hz	F#2 / Gb2	89.86 – 95.21 Hz	F#4 / Gb4	359.45 – 380.83 Hz
G0	23.8 – 25.21 Hz	G2	95.21 – 100.87 Hz	G4	380.84 – 403.48 Hz
G#0 / Ab0	25.22 – 26.72 Hz	G#2 / Ab2	100.87 – 106.87 Hz	G#4 / Ab4	403.47 – 427.46 Hz
A0	26.71 – 28.3 Hz	A2	106.86 – 113.22 Hz	A4	427.47 – 452.89 Hz
A#0 / Bb0	28.31 – 29.99 Hz	A#2 / Bb2	113.22 – 119.95 Hz	A#4 / Bb4	452.88 – 479.81 Hz
B0	29.99 – 31.77 Hz	B2	119.95 – 127.08 Hz	B4	479.82 – 508.35 Hz
C1	31.76 – 33.65 Hz	C3	127.08 – 134.64 Hz	C5	508.35 – 538.58 Hz
C#1 / Db1	33.66 – 35.66 Hz	C#3 / Db3	134.64 – 142.65 Hz	C#5 / Db5	538.58 – 570.61 Hz
D1	35.66 – 37.78 Hz	D3	142.65 – 151.13 Hz	D5	570.6 – 604.53 Hz
D#1 / Eb1	37.78 – 40.02 Hz	D#3 / Eb3	151.13 – 160.11 Hz	D#5 / Eb5	604.53 – 640.48 Hz
E1	40.02 – 42.4 Hz	E3	160.11 – 169.63 Hz	E5	640.49 – 678.57 Hz
F1	42.4 – 44.92 Hz	F3	169.63 – 179.72 Hz	F5	678.57 – 718.92 Hz
F#1 / Gb1	44.93 – 47.6 Hz	F#3 / Gb3	179.73 – 190.42 Hz	F#5 / Gb5	718.92 – 761.67 Hz
G1	47.6 – 50.43 Hz	G3	190.42 – 201.74 Hz	G5	761.67 – 806.96 Hz
G#1 / Ab1	50.43 – 53.43 Hz	G#3 / Ab3	201.73 – 213.73 Hz	G#5 / Ab5	806.96 – 854.94 Hz
A1	53.43 – 56.61 Hz	A3	213.73 – 226.44 Hz	A5	854.94 – 905.78 Hz
A#1 / Bb1	56.61 – 59.97 Hz	A#3 / Bb3	226.44 – 239.9 Hz	A#5 / Bb5	905.78 – 959.64 Hz
B1	59.98 – 63.54 Hz	B3	239.91 – 254.17 Hz	B5	959.65 – 1016.71 Hz

C6	1016.7 – 1077.16 Hz
C#6 / Db6	1077.16 – 1141.21 Hz
D6	1141.21 – 1209.07 Hz
D#6 / Eb6	1209.08 – 1280.97 Hz
E6	1280.97 – 1357.14 Hz
F6	1357.14 – 1437.84 Hz
F#6 / Gb6	1437.84 – 1523.34 Hz
G6	1523.34 – 1613.92 Hz
G#6 / Ab6	1613.92 – 1709.89 Hz
A6	1709.89 – 1811.57 Hz
A#6 / Bb6	1811.57 – 1919.29 Hz
B6	1919.29 – 2033.41 Hz

C7	2033.41 – 2154.32 Hz
C#7 / Db7	2154.33 – 2282.43 Hz
D7	2282.43 – 2418.15 Hz
D#7 / Eb7	2418.16 – 2561.95 Hz
E7	2561.94 – 2714.28 Hz
F7	2714.29 – 2875.69 Hz
F#7 / Gb7	2875.69 – 3046.69 Hz
G7	3046.68 – 3227.84 Hz
G#7 / Ab7	3227.85 – 3419.79 Hz
A7	3419.79 – 3623.14 Hz
A#7 / Bb7	3623.14 – 3838.58 Hz
B7	3838.59 – 4066.84 Hz

C8	4066.84 – 4308.66 Hz
C#8 / Db8	4308.66 – 4564.87 Hz
D8	4564.87 – 4836.31 Hz
D#8 / Eb8	4836.31 – 5123.89 Hz

8.4 Thomae's Function

Thomae's function is a mathematical function that is 0 at the irrational numbers and reciprocal of the denominator at rational numbers:

$$\tau(t) = \begin{cases} \frac{1}{q} & \text{if } t = \frac{p}{q} \text{ (} t \text{ is rational), with } p \in \mathbb{Z} \text{ and } q \in \mathbb{N} \text{ coprime} \\ 0 & \text{if } t \text{ is irrational} \end{cases} \quad 8.1$$

Below is the graph of Thomae's function

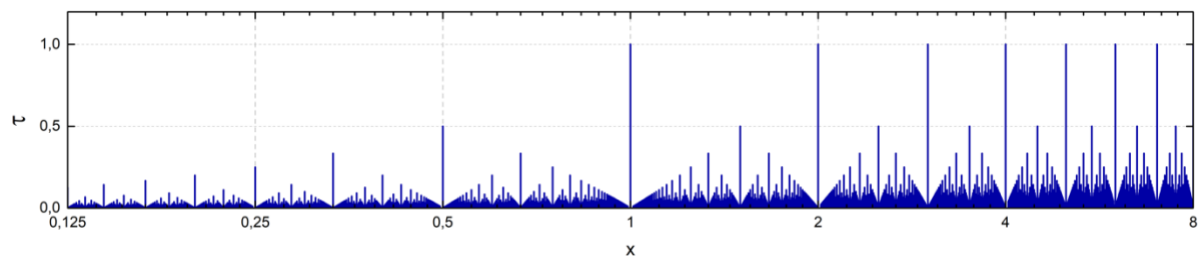


Figure 8.1 - Thomae's function as described by the equation 8.1. Note that for $x < 1$ the graph is identical to Figure 5.5.

8.5 Proof of harmonicity as inverse of interval's complexity for single frequency sounds

Here we are going to prove that harmonicity distribution can be described by Formula 5.9 when contextual and complementary sets have a single frequency $F = F' = f\mathbb{N}_1$. To demonstrate that, let's use Formula 5.6 to find a total consonance when complementary set is transposed on an arbitrary interval t :

$$\sigma(f\mathbb{N}_1, tf\mathbb{N}_1) = \frac{\alpha(f\mathbb{N}_1, tf\mathbb{N}_1)}{2} + \frac{\chi(f\mathbb{N}_1, tf\mathbb{N}_1)}{2} \quad 8.2$$

If t is irrational, then both affinity and harmonicity will equal to 0, just like in the Formula 5.9. In the rational case, when $t = \frac{p}{q}$ with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime, we get the following formula for total consonance:

$$\sigma\left(f\mathbb{N}_1, \frac{p}{q}f\mathbb{N}_1\right) = \frac{|f\mathbb{N}_1 \cap \frac{p}{q}f\mathbb{N}_1|}{2 \cdot \min(|f\mathbb{N}_1|, |\frac{p}{q}f\mathbb{N}_1|)} + \frac{\gcd(f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1) \cdot |f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1|}{2 \cdot \max(f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1)} \quad 8.3$$

In our case, when both sets in the arguments have only 1 partial, it is always true that $\min(|f\mathbb{N}_1|, |\frac{p}{q}f\mathbb{N}_1|) = 1$. Also, the value f can be factored out from $\gcd(f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1) = f \cdot \gcd(1, \frac{p}{q})$ and $\max(f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1) = f \cdot \max(1, \frac{p}{q})$. Given that, we can simplify our formula of total consonance:

$$\sigma\left(f\mathbb{N}_1, \frac{p}{q}f\mathbb{N}_1\right) = \frac{|f\mathbb{N}_1 \cap \frac{p}{q}f\mathbb{N}_1|}{2} + \frac{\gcd(1, \frac{p}{q}) \cdot |f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1|}{2 \cdot \max(1, \frac{p}{q})} \quad 8.4$$

One can notice that $\gcd(1, \frac{p}{q}) = \frac{1}{q}$ and factoring that additional q in the denominator into the $\max(1, \frac{p}{q})$ gives us the following:

$$\sigma\left(f\mathbb{N}_1, \frac{p}{q}f\mathbb{N}_1\right) = \frac{|f\mathbb{N}_1 \cap \frac{p}{q}f\mathbb{N}_1|}{2} + \frac{|f\mathbb{N}_1 \cup \frac{p}{q}f\mathbb{N}_1|}{2 \cdot \max(p, q)} \quad 8.5$$

Now let's look at 2 cases when $\frac{p}{q} = 1$ and when $\frac{p}{q} \neq 1$. In the first case, the requirement of p and q being coprime leads to $p = q = 1$ and thus we can write:

$$\frac{p}{q} = 1: \quad \sigma(f\mathbb{N}_1, f\mathbb{N}_1) = \frac{1}{2} + \frac{1}{2 \cdot 1} = 1 \quad 8.6$$

In the second case, all of the contributions from set sizes have definitive values and the formula can be simplified:

$$\frac{p}{q} \neq 1: \quad \sigma\left(f\mathbb{N}_1, \frac{p}{q}f\mathbb{N}_1\right) = \frac{0}{2} + \frac{2}{2 \cdot \max(p, q)} = \frac{1}{\max(p, q)} \quad 8.7$$

However, one can notice, that $\frac{1}{\max(p, q)} = 1$ if $p = q = 1$. Thus, the formula for the second case $\frac{p}{q} \neq 1$ can be extended for the case of $\frac{p}{q} = 1$ as it has the same value. Combining all the cases into the single formula we can write:

$$\sigma(f\mathbb{N}_1, tf\mathbb{N}_1) = \begin{cases} \frac{1}{\max(p, q)} & \text{if } t = \frac{p}{q} \text{ is rational} \\ 0 & \text{if } t \text{ is irrational} \end{cases} \quad 8.8$$

Which is the same as Formula 5.9, as was to be proved!

About the New Tonality

New Tonality started as a YouTube channel in 2020. The purpose was to share the scientific theories behind the nature of musical consonance and dissonance. The current ambition is to bring those ideas to practice for musicians around the world. The author's background in physics, software development and music places him in the unique position to develop such tools, some of which can be found on the newtonality.net website.

To follow the updates, subscribe to the New Tonality [YouTube channel](#) or become a follower on [Ko-Fi](#). To get a glimpse of what music can be made by implementing the ideas discussed in the eBook, check out [Offcast](#) on Bandcamp.

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